UNIT 2: FOOD MICROBIOLOGY, SPOILAGE AND PRESERVATION — MASTER STUDY MANUAL

Part I: Module 1 — Food-borne Bacteria: Taxonomy, Physiology and Pathogenesis

Evidence status and reading convention

The supplied search result contains generic research websites—not OUAT Lessons 18–26, an official HPSC paper, or the relevant statutory schedules. It therefore cannot substantiate quotations from the course, the "latest HPSC pattern," or current legal limits.

Accordingly:

- This is an **original technical study manual**, not a verified verbatim reproduction of OUAT lessons.
- Statements attributed to the course in your request are identified as **reported historical wording, pending verification**.
- Modern scientific corrections are presented separately.
- The assessment sections use your requested count-based and conventional formats; their adoption in a particular HPSC examination has not been independently verified.
- A microbiological growth boundary is **not automatically a statutory processing standard**.

ESSENTIAL CORRECTION LEDGER

The ledger distinguishes what a historical source reportedly says from what should govern a scientifically correct answer.

Issue	Historical/course-side evidence status	Scientifically correct interpretation
UHT stated as **93.4°C for 3 s**	Reported in the request; original lesson unavailable	This is not an adequate general definition of UHT treatment for commercially sterile milk. UHT commonly operates around **135–150°C for a few seconds**, with a validated process and aseptic packaging.
Blanket statements that sulphites are "banned"	Reported historical overgeneralisation; precise wording unavailable	Permission, prohibition, maximum level and declaration requirements depend on jurisdiction, food category and current regulation. Neither "permitted everywhere" nor "banned everywhere" is correct.
Natamycin surface limit written in **mg/dm³**	Reported dimensional error; original table unavailable	Surface loading is expressed per **area**, such as mg/dm², where the applicable regulation specifies a surface limit. A concentration per volume is a different quantity.
Irradiation source written as **Ce-137**	Reported misprint; original text unavailable	The relevant radionuclide is **caesium-137, Cs-137**. "Ce" denotes cerium.
Ordinary food poisoning caused by *C.*	Historical toxinotyping terminology	Many classical enterotoxigenic type A

perfringens “type A”		food-poisoning strains are designated **type F** under the expanded A–G toxinotyping scheme. Older literature remains important for examination interpretation.
Quick freezing invariably produces intracellular ice	Simplification in the requested blueprint, not verified as an OUAT quotation	Rapid freezing generally produces smaller crystals and less structural damage, but **ice location is not fixed solely by the words “quick” and “slow.”** Intracellular ice can itself be highly damaging.
“All *Pseudomonas* are oxidase-positive”	Common teaching generalisation	Most familiar food-associated pseudomonads are oxidase-positive, but exceptions exist. Identification requires a test panel.
“Seven botulinum toxin types exhaust all modern diversity”	Classical classification	A–G remain the standard classical antigenic types. Modern work also recognises mosaic and additional toxin diversity.
All lactic acid bacteria are beneficial, non-pathogenic and catalase-negative	Common categorical simplification	LAB are a functional group with strain-level variation. Some have catalase or pseudocatalase activity under particular conditions; some can cause spoilage, form biogenic amines or act as opportunistic pathogens.

Dual-evidence rule: When an examination explicitly asks for historical course terminology, identify that terminology, then add the modern correction where the answer format permits. Do not silently convert an unverified historical quotation into established fact.

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1. FIRST-PRINCIPLES MODEL: WHY A BACTERIUM GROWS IN A FOOD

A food is simultaneously:

1. A source of carbon, nitrogen, minerals and growth factors.
2. An aqueous environment with a particular water activity.
3. A chemical environment defined by pH, buffering capacity and antimicrobial compounds.
4. A redox environment influenced by oxygen diffusion, packaging and microbial metabolism.
5. A physical structure containing surfaces, pores, fat droplets, particles and protected microenvironments.

Microbial behaviour follows from the interaction:

$$
[
\text{Organism phenotype}
\times
\text{Food environment}
\times
\text{Time–temperature history}
]
$$

A species name alone does not determine risk.

For example:

- *Salmonella* may survive in a dry food without multiplying.
- *Listeria monocytogenes* may multiply slowly during refrigeration.
- *Clostridium perfringens* spores may survive cooking and subsequently germinate during slow cooling.

- *Staphylococcus aureus* cells may be killed by heating after enterotoxin has already accumulated.
- *Bacillus cereus* spores may survive cooking, while the outcome depends on whether the strain produces enterotoxins or cereulide.

1.1 Five events that must not be confused

Event	Meaning	Food-safety consequence
Contamination	Organism or toxin enters the food	Establishes a possible hazard, not necessarily growth
Survival	Organism remains viable	Can occur without multiplication
Growth	Cell number increases	May increase infectious dose or permit toxin accumulation
Toxin production	A strain expresses and releases a toxin	Depends on strain, substrate and environment
Inactivation	Viability or toxin activity decreases	Killing cells does not necessarily destroy their toxins

> ***[HPSC EXAM TRAP]***
> *"The organism cannot grow in this food" does* ***not*** *mean "the organism cannot survive in this food."*

1.2 Infection, intoxication and toxicoinfection

A. Food-borne infection

Viable organisms are ingested and establish infection or colonisation.

Examples:

- *Salmonella enterica*
- *Listeria monocytogenes*
- EIEC
- STEC

Some infections are toxin-mediated after ingestion. STEC remains an infection even though Shiga toxin is central to disease.

B. Food-borne intoxication

A toxin already present in the food is ingested.

Examples:

- Staphylococcal enterotoxin poisoning.
- Emetic *B. cereus* poisoning.
- Classical food-borne botulism.

C. Toxicoinfection

Viable organisms are ingested, followed by toxin production in the intestinal tract.

Examples:

- Classical *C. perfringens* food poisoning.
- The principal model of diarrhoeal *B. cereus* disease.

> ***[HPSC EXAM TRAP]***
> *Do not classify a disease merely because the organism "produces toxin." Ask* ***where and when the clinically important toxin is produced****.*

2. FOUNDATIONAL BACTERIAL PHYSIOLOGY

2.1 Gram reaction: a cell-envelope phenomenon

Mental model

The Gram stain is a differential staining procedure reflecting how the cell envelope retains a crystal-violet–iodine complex during decolourisation. It is not a direct assay of pathogenicity, spore formation or oxygen requirement.

Procedure and mechanism

Step	Reagent	Principal event
1	Crystal violet	Initially stains both major groups
2	Iodine	Forms a crystal-violet–iodine complex
3	Alcohol/acetone decolouriser	Produces differential retention
4	Counterstain, commonly safranin	Colours decolourised cells pink/red

Gram-positive envelope

- Thick peptidoglycan.
- Usually contains teichoic and lipoteichoic acids.
- Lacks the characteristic Gram-negative outer membrane.
- Retains the primary stain under correctly controlled conditions.

Gram-negative envelope

- Relatively thin peptidoglycan.
- An outer membrane containing lipopolysaccharide in typical Gram-negative bacteria.
- A periplasmic compartment.
- Decolourisation permits subsequent counterstaining.

Why errors occur

- Old cultures may become Gram-variable.
- Excessive decolourisation can make Gram-positive organisms appear Gram-negative.
- Insufficient decolourisation can produce the opposite error.
- Damaged cells may stain atypically.
- Food debris can obscure morphology.

> ***[HPSC EXAM TRAP]***
> *Gram-positive does not mean "spore-forming." Listeria, Staphylococcus, Carnobacterium and Corynebacterium do not form the bacterial endospores characteristic of Bacillus and Clostridium.*

2.2 Catalase: removal of hydrogen peroxide

Mental model

Oxygen metabolism can generate reactive oxygen species. Catalase removes hydrogen peroxide:

[
2H_2O_2 \xrightarrow{\text{catalase}} 2H_2O + O_2\uparrow
]

Visible bubbling constitutes a positive test under appropriate conditions.

Major examination associations

Organism/group	Typical catalase reaction
Staphylococcus	Positive
Listeria	Positive
Corynebacterium	Positive
Bacillus	Usually positive
Acinetobacter	Positive
Many pseudomonads	Positive
Most conventional LAB	Negative

Limitations

- "Catalase-negative" does not mean "strict anaerobe."
- Blood carried over from blood agar can produce a misleading reaction.
- Some LAB possess heme-dependent catalase or manganese-associated pseudocatalase activity.
- Catalase testing does not establish whether an organism can cause food poisoning.

> ***[HPSC EXAM TRAP]***
> *Catalase decomposes **hydrogen peroxide**, not molecular oxygen. The oxygen bubbles are a reaction product.*

2.3 Oxidase: a respiratory-chain test

Mental model

The oxidase test detects activity associated with certain cytochrome-c-dependent respiratory systems through oxidation of a reagent.

It does not simply ask whether a bacterium uses oxygen.

Typical differentiation

Organism/group	Typical oxidase reaction
Acinetobacter	Negative
Most familiar *Pseudomonas* species	Positive
Escherichia coli	Negative
Salmonella	Negative
Listeria	Negative

A strict aerobe can be oxidase-negative because its respiratory chain need not contain the particular enzyme system detected by the test.

> ***[HPSC EXAM TRAP]***
> *"Aerobic" and "oxidase-positive" are not synonyms. Acinetobacter is the standard counterexample.*

2.4 Atmospheric niches

Mental model

Oxygen has two opposing roles:

- It is an efficient terminal electron acceptor.
- It can generate damaging reactive oxygen species.

The growth response therefore depends on energy metabolism and oxidative-stress defences.

Category	Physiological meaning	Relevant examples
Strict/obligate aerobe	Requires oxygen for ordinary growth	*Acinetobacter*; many food-spoilage pseudomonads
Microaerophile	Grows best at oxygen concentrations below atmospheric levels	*Campylobacter* is the standard food-safety example
Facultative anaerobe	Can grow with or without oxygen using different metabolic routes	*E. coli*, *Salmonella*, *Listeria*, *S. aureus*
Aerotolerant anaerobe	Does not principally depend on oxygen for energy generation but tolerates exposure	Many LAB
Obligate anaerobe	Growth requires sufficiently anaerobic conditions	*C. botulinum*, *C. perfringens*

Important qualifications

6. Some *Pseudomonas* species can grow anaerobically using alternative electron acceptors such as nitrate.
7. Facultative anaerobes may grow faster aerobically because respiration yields more energy than fermentation.
8. Obligate anaerobes vary in tolerance of brief oxygen exposure.
9. The atmosphere surrounding a package does not necessarily represent the redox state inside a food particle.

> ***[HPSC EXAM TRAP]***
> *Vacuum packaging suppresses many aerobic spoilage organisms but does not make food sterile. It may select for LAB, facultative anaerobes and, when other controls fail, anaerobic pathogens.*

2.5 Redox potential and microbial succession

Redox potential, (E_h), expresses the tendency of a system to accept or donate electrons.

- More oxidising environments tend to favour organisms relying on aerobic respiration.
- Reduced environments favour organisms adapted to low-oxygen metabolism.
- Microbial oxygen consumption can lower local (E_h).
- Protein-rich food interiors and tightly packed particles may contain reduced microenvironments.

A succession can occur:

[
\text{Initial aerobic growth}
\rightarrow
\text{oxygen depletion}
\rightarrow

\text{selection of facultative/anaerobic organisms}
]

This is a tendency, not a universal sequence.

2.6 Endospores versus vegetative cells

Mental model

An endospore is a survival structure, not a reproductive structure.

A single vegetative cell generally forms one endospore. Germination ordinarily gives rise to one vegetative cell.

Relevant genera

- *Bacillus* and related genera.
- *Clostridium* and related genera.

Resistance mechanisms

Endospore resistance reflects several interacting features:

- A dehydrated core.
- Calcium–dipicolinate.
- Small acid-soluble spore proteins protecting DNA.
- Cortex and coat structures.
- Reduced metabolic activity.
- Repair mechanisms operating during return to growth.

Developmental sequence

[
\text{Vegetative cell}
\xrightarrow[\text{stress/signalling}]{\text{sporulation}}
\text{endospore}
]

[
\text{Endospore}
\xrightarrow{\text{germination}}
\text{germinated spore}
\xrightarrow{\text{outgrowth}}
\text{vegetative cell}
]

Distinctions

Feature	Vegetative cell	Endospore
Metabolic activity	Active under permissive conditions	Very low/dormant
Multiplication	By cell division	Does not multiply as a spore
Heat resistance	Usually lower	Usually substantially higher
Drying resistance	Variable	Often high
Function	Growth and metabolism	Persistence and survival

Heat activation is not sterilisation

Sublethal heating may activate some spores, making subsequent germination more likely under suitable conditions. The same treatment may kill competing vegetative organisms.

Thus:

[
\text{Cooking}
+
\text{spore survival}
+
\text{slow cooling}
=
\text{potential post-cooking hazard}
]

> ***[HPSC EXAM TRAP]***
> *Sporulation is not bacterial multiplication. "One cell becomes many spores" is incorrect for ordinary bacterial endospore formation.*

2.7 Growth mathematics: why a short delay matters

During ideal exponential growth:

[
N_t=N_0\,2^{t/g}
]

where:

- (N_t): population after time (t);
- (N_0): initial population;
- (g): generation time.

The specific growth rate, (\mu), is related to generation time by:

[
\mu=\frac{\ln 2}{g}
]

and:

[
N_t=N_0e^{\mu t}
]

Example

If a population has an 8-minute generation time under favourable conditions:

[
n=\frac{80}{8}=10\text{ generations}
]

[
N{80}=N_0\,2^{10}=1024N_0
]

This does **not** imply that all food populations maintain an 8-minute generation time. Lag, nutrient limitation, competing flora and temperature variation invalidate unrestricted extrapolation.

> ***[HPSC EXAM TRAP]***
> *Minimum generation time is an optimum-condition measurement—not a constant applicable throughout refrigeration, cooling or storage.*

3. SYSTEMATIC BACTERIAL PROFILES

3.1 *Acinetobacter* versus *Pseudomonas*

First-principles model: oxygen-driven spoilage on exposed surfaces

Chilled foods often suppress mesophiles without eliminating organisms capable of growth at low temperatures. On oxygenated surfaces, respiratory organisms can consume readily available substrates and subsequently attack proteins or lipids.

Typical spoilage effects include:

- Slime.
- Off-odours.
- Pigmentation.
- Proteolysis.
- Lipolysis.

Comparative identification matrix

Feature	*Acinetobacter*	*Pseudomonas*
Gram reaction	Negative; sometimes appears Gram-variable	Negative
Morphology	Short rods/coccobacilli; may appear coccoid	Rods
Catalase	Positive	Usually positive
Oxidase	Negative	Usually positive; exceptions occur
Conventional flagellar motility	Absent	Commonly present, often polar flagella
Oxygen relationship	Generally obligately aerobic	Generally respiratory/aerobic; some species can use alternative electron acceptors
Food relevance	Spoilage-associated environmental flora	Major aerobic spoilage organisms in chilled foods
Spore formation	Absent	Absent

Acinetobacter

Relevant features:

- Found in soil, water and food-processing environments.
- Non-flagellated in routine identification.
- Some strains show surface-associated movement; this should not be confused with ordinary flagellar motility.
- Can contribute to chilled-food spoilage, depending on species and product.
- Clinical importance of some species does not make the genus a standard cause of classical food intoxication.

Pseudomonas

Important food-associated species include:

- *P. fluorescens* group.
- *P. fragi*.
- *P. lundensis*.

In milk and meat:

10. Psychrotrophic strains multiply during chilled storage.
11. Extracellular proteases and lipases may accumulate.
12. A later heat treatment may kill the cells without fully inactivating all previously formed enzymes.
13. Continued enzyme action can cause bitterness, rancidity or physical instability.

Cells and extracellular enzymes have different thermal sensitivities.

> ***[HPSC EXAM TRAP]***
> *- Acinetobacter: oxidase-negative and conventionally non-motile.*
> *- Pseudomonas: usually oxidase-positive and motile.*
> *- Neither genus is defined by endospore formation.*
> *- "All pseudomonads are strict aerobes under every condition" is too absolute.*

3.2 *Bacillus cereus*

First-principles model: a heat-surviving spore followed by two different toxin pathways

Cooking may destroy vegetative cells but leave spores. The safety outcome depends on subsequent time–temperature conditions and the toxin phenotype of the strain.

Two distinct disease mechanisms must be separated:

[
\text{Diarrhoeal disease}
\neq
\text{Emetic disease}
]

Basic profile

- Gram-positive rods, although older cultures may be Gram-variable.
- Endospore-forming.
- Usually motile.
- Catalase-positive.
- Aerobic or facultatively anaerobic.
- Member of the *B. cereus* sensu lato group.
- Growth range is strain-dependent; some members are psychrotolerant.

A food isolate identified only as belonging to the *B. cereus* group is not automatically an emetic strain.

A. Diarrhoeal syndrome

Principal toxins

- Haemolysin BL, **Hbl**.
- Non-haemolytic enterotoxin, **Nhe**.
- Cytotoxin K, **CytK**.

Hbl and Nhe are multicomponent pore-forming toxin systems. CytK is also associated with membrane injury.

Pathway

[
\text{Contaminated food containing viable cells/spores}
\rightarrow
\text{survival through ingestion}
\rightarrow
\text{growth/toxin production in intestine}
\rightarrow
\text{epithelial injury and diarrhoea}
]

Clinical pattern

- Incubation commonly approximately **6–15 hours**, with wider reported ranges.
- Watery diarrhoea.
- Abdominal cramps.
- Vomiting is less characteristic than in emetic illness.

Vehicles

- Meat dishes.
- Sauces.
- Vegetables.
- Dairy products.
- Mixed foods.

The diarrhoeal enterotoxins are proteins and are generally more heat-labile than cereulide.

B. Emetic syndrome

Toxin identity

Cereulide is:

- A cyclic dodecadepsipeptide.
- Produced by non-ribosomal peptide synthesis.
- Encoded through the cereulide synthetase system in emetic strains.
- Relatively resistant to heat, acid and digestive proteolysis.
- A potassium ionophore capable of disrupting mitochondrial function.

A depsipeptide contains both peptide and ester linkages.

Pathway

[
\text{Growth of an emetic strain in food}
\rightarrow
\text{preformed cereulide}
\rightarrow
\text{ingestion}
\rightarrow
\text{rapid nausea and vomiting}
]

Clinical pattern

- Incubation commonly about **0.5–6 hours**.
- Nausea and vomiting predominate.
- Severe cases can involve systemic toxicity.

Typical vehicles

- Cooked rice.
- Pasta.
- Other starch-rich prepared foods.

Rice is a classical association, not an exclusive vehicle.

Comparative matrix

Feature	Diarrhoeal syndrome	Emetic syndrome
Main hazardous agents	Hbl, Nhe, CytK	Cereulide
Chemical nature	Proteins	Cyclic depsipeptide
Principal toxin-production site	Intestine in the classical disease model	Food before ingestion
Disease category	Toxicoinfection	Intoxication
Typical onset	Later	Earlier
Dominant symptom	Diarrhoea	Vomiting
Heat stability	Generally relatively heat-labile	Relatively heat-stable
Classical vehicle	Mixed dishes, meats, sauces	Cooked rice and starch-rich foods

Control logic

14. Prevent extensive post-cooking growth.
15. Cool cooked food promptly using a validated procedure.
16. Maintain appropriate hot or cold storage.
17. Avoid treating reheating as a remedy for previously accumulated cereulide.
18. Consider psychrotolerant strains in chilled products.

> ***[HPSC EXAM TRAP]***
> *- Cereulide is **not** the same toxin as Nhe or Hbl.*
> *- Killing B. cereus cells does not guarantee destruction of cereulide.*
> *- All B. cereus strains do not produce cereulide.*
> *- The organism's spores, vegetative cells and toxins require separate control considerations.*

3.3 *Bacillus subtilis*

First-principles model: extracellular enzymes convert food structure into nutrients

A bacterium can spoil food through enzymes even when its abundance is not the only determinant of quality loss.

B. subtilis and related bacilli secrete enzymes that hydrolyse proteins and carbohydrates. In bread, surviving spores can germinate when moisture, temperature and acidity are favourable.

Basic profile

- Gram-positive rod.

- Endospore-forming.
- Usually motile.
- Catalase-positive.
- Principally aerobic, with anaerobic metabolic capabilities in suitable strains and conditions.

Subtilisin

Subtilisin is an extracellular **serine protease**.

Its catalytic machinery includes a:

[
\text{Asp–His–Ser catalytic triad}
]

The enzyme hydrolyses peptide bonds:

[
\text{Protein}+H_2O
\xrightarrow{\text{protease}}
\text{peptides/amino acids}
]

Industrial uses include enzyme production, but usefulness of selected strains does not eliminate the spoilage potential of uncontrolled growth in foods.

Rope spoilage of bread

Sequence

19. Spores enter flour or dough.
20. Some survive baking, particularly within the moist crumb where the temperature history differs from the oven air.
21. Bread cools slowly or is stored under warm, humid conditions.
22. Spores germinate and vegetative cells grow.
23. Enzymes and extracellular polymers contribute to crumb breakdown and ropiness.

Manifestations

- Sticky or stringy crumb.
- Brownish discolouration.
- Unpleasant, sometimes fruity odour.
- Loss of normal crumb structure.

Biochemical contributors

- Amylolytic activity.
- Proteolytic activity.
- Extracellular polysaccharides.

Some bacilli can produce **levan**, a fructan formed from sucrose by levansucrase. Rope spoilage should not be attributed to one universal polymer or to *B. subtilis* alone; related bacilli also participate.

Control logic

- Raw-material hygiene.
- Prompt cooling.
- Appropriate storage.
- Acidification or approved antimycotic/antibacterial preservation where applicable.

- Prevention of equipment and environmental contamination.

> ***[HPSC EXAM TRAP]***
> *- Baking temperature is not identical to the temperature reached throughout the bread crumb.*
> *- Rope spoilage is not necessarily mould spoilage.*
> *- "Subtilisin" is a protease, not a polysaccharide.*
> *- Levan is a fructan; it should not be casually equated with dextran.*

3.4 *Carnobacterium*

First-principles model: refrigeration and vacuum packaging alter competition

Removing oxygen suppresses many aerobic competitors. Refrigeration then selects organisms that can function at low temperatures. *Carnobacterium* can occupy this niche in meats and seafood.

Basic profile

- Gram-positive rods.
- Non-spore-forming.
- Generally catalase-negative in routine testing.
- Fermentative LAB.
- Facultatively anaerobic/aerotolerant growth characteristics.
- Psychrotrophic species and strains.
- Able to occur in vacuum- or modified-atmosphere-packaged foods.

Important species include:

- *Carnobacterium divergens*.
- *Carnobacterium maltaromaticum*, historically associated with the name *C. piscicola*.

Fermentation products vary with species, substrate and conditions; genus membership alone should not be used to infer one invariant metabolic pattern.

A. Spoilage role

Possible consequences include:

- Off-odours.
- Changes in flavour.
- Metabolite accumulation.
- Product-specific sensory deterioration.

Their presence does not always imply obvious spoilage.

B. Protective-culture role

Some strains produce bacteriocins active against susceptible *Listeria* strains.

Examples include:

- Carnobacteriocins.
- Divercin V41.
- Piscicolin 126.

Several are class IIa, pediocin-like bacteriocins. Their action commonly involves interaction with a susceptible-cell membrane system and membrane permeabilisation.

Why "anti-listerial" does not mean "universally safe"

A protective strain must be assessed for:

- Effective inhibition in the actual food.
- Stability during storage.
- Sensory effects.
- Bacteriocin resistance.
- Biogenic-amine production.
- Other strain-level safety characteristics.

C. Tyramine formation

Some strains decarboxylate tyrosine:

[
\text{L-tyrosine}
\xrightarrow{\text{tyrosine decarboxylase}}
\text{tyramine}+CO_2
]

A more mechanistically explicit representation includes proton consumption:

[
\text{Tyrosine}+H^+
\rightarrow
\text{Tyramine}+CO_2
]

Amino-acid decarboxylation can contribute to acid-stress adaptation. Tyramine accumulation is undesirable, particularly for susceptible consumers and those taking certain monoamine-oxidase-inhibiting medicines.

> ***[HPSC EXAM TRAP]***
> *- A bacteriocin-producing strain is not automatically a suitable starter.*
> *- Anti-listerial activity and tyramine production can coexist.*
> *- Vacuum packaging does not necessarily inhibit Carnobacterium.*

3.5 *Corynebacterium*

First-principles model: the same broad taxonomic group can contain industrial producers and pathogens

Food microbiology must distinguish:

- Genus-level morphology.
- Species-level ecology.
- Strain-level toxigenicity.

The industrial role of *C. glutamicum* does not imply that all corynebacteria are beneficial.

General morphology and physiology

- Gram-positive, non-spore-forming rods.
- Often pleomorphic or club-shaped.
- “Chinese-letter” or palisade arrangements can arise through snapping division.
- Usually catalase-positive.
- Some species contain mycolic acids in their cell envelope.
- High-GC actinobacterial lineage.

A. *Corynebacterium glutamicum*: amino-acid production

Glutamate biosynthesis

Carbon flows through glycolysis and the tricarboxylic acid pathway toward α-ketoglutarate.

A simplified reaction is:

[
\alpha\text{-ketoglutarate}+NH_4^++NADPH
\rightarrow
L\text{-glutamate}+NADP^++H_2O
]

Glutamate dehydrogenase provides a major route under suitable production conditions.

Lysine biosynthesis

A simplified precursor pathway is:

[
\text{Oxaloacetate}
\rightarrow
\text{Aspartate}
\rightarrow
\text{Aspartate-family intermediates}
\rightarrow
\text{Diaminopimelate}
\rightarrow
L\text{-lysine}
]

Industrial principles

- Redirect carbon flux toward product.
- Provide adequate reducing power.
- Control nitrogen supply.
- Reduce feedback inhibition through strain selection or engineering.
- Promote export.

Glutamate release is associated with changes in membrane physiology and export systems, including mechanosensitive-channel-mediated export. It should not be explained simply as “cells burst and release glutamate.”

B. *Corynebacterium diphtheriae*: A–B toxin architecture

C. diphtheriae is primarily a pathogen of respiratory or cutaneous infection, not a standard cause of classical food-borne intoxication. Its inclusion is valuable for bacterial physiology and toxin-mechanism questions.

Toxigenicity

- The toxin gene is associated with a toxigenic bacteriophage.
- Not all *C. diphtheriae* isolates produce diphtheria toxin.
- Iron availability regulates toxin expression through the DtxR regulatory system.

In simplified terms:

- Sufficient iron supports repression.
- Iron limitation can permit toxin expression in a toxigenic strain.

Structure

Diphtheria toxin is produced as a polypeptide that can be proteolytically nicked into functional fragments held together by a disulphide linkage.

Fragment	Principal function
A	Catalytic ADP-ribosyltransferase
B	Receptor binding and translocation functions

Stepwise mechanism

24. Fragment B binds the appropriate host-cell receptor.
25. The toxin enters by receptor-mediated endocytosis.
26. Endosomal acidification promotes membrane insertion/translocation.
27. Fragment A reaches the cytosol.
28. Fragment A uses NAD^+ to ADP-ribosylate elongation factor 2.
29. Protein synthesis is inhibited.

$$
\text{NAD}^+ + \text{eEF2} \xrightarrow{\text{fragment A}} \text{ADP-ribosyl-eEF2} + \text{nicotinamide}
$$

The modified site is the diphthamide residue of eEF2.

> ***[HPSC EXAM TRAP]***
> *- **A** = **active/catalytic; B** = **binding/translocation.***
> *- Diphtheria toxin modifies **eEF2**, not bacterial EF-Tu.*
> *- It does not act by the same ribosomal depurination mechanism as Shiga toxin.*
> *- The presence of a species does not establish the presence or expression of its toxin gene.*

3.6 *Clostridium perfringens*

First-principles model: cooking selects spores, while slow cooling permits explosive growth

Large meat masses cool unevenly. The exterior may cool while the interior remains within a favourable growth range. Cooking can remove competitors while leaving spores able to germinate.

This creates the sequence:

$$
\text{Spore survival} \rightarrow \text{germination} \rightarrow \text{rapid vegetative growth} \rightarrow \text{ingestion} \rightarrow \text{intestinal sporulation and enterotoxin production}
$$

Basic profile

- Gram-positive, large rods.

- Endospore-forming.
- Anaerobic.
- Relatively aerotolerant compared with some strict anaerobes.
- Conventionally non-motile in routine identification.
- Often exhibits a double zone of haemolysis.
- Produces lecithinase associated with alpha toxin.

Absence of flagella does not exclude every form of surface-associated movement.

Growth kinetics

Under favourable conditions, generation times may be **less than 10 minutes**.

Growth is especially rapid at warm temperatures near the upper part of its mesophilic range, commonly around the low-to-mid 40s °C, depending on strain and medium.

This is why warm holding and slow cooling of meat dishes are critical.

Vehicles

- Cooked meat.
- Poultry.
- Gravy.
- Stews.
- Bulk catering dishes.

The combination of substantial food mass, surviving spores and poor cooling is more important than a particular recipe name.

A. Historical type A versus modern type F

Older food-microbiology texts identify ordinary food poisoning with **type A** strains.

The expanded toxinotyping system separates enterotoxigenic strains producing alpha toxin and CPE, but lacking the defining beta, epsilon and iota toxins, into **type F**.

Thus an examination-safe statement is:

> *Classical C. perfringens food poisoning was historically associated with enterotoxigenic type A strains; many such strains are classified as type F in the expanded scheme.*

Not every historical type A strain carries the enterotoxin gene.

B. Enterotoxin: CPE

CPE is a protein of approximately 35 kDa.

Production and release

- Expression is strongly associated with sporulation.
- In classical food poisoning, sporulation occurs in the intestine.
- Toxin is released as the mother cell lyses.

Cellular action

30. CPE binds susceptible claudin receptors.
31. Toxin–receptor complexes assemble.
32. Membrane pores form.

33. Ionic balance is disrupted.
34. Epithelial injury and altered intestinal fluid handling produce diarrhoea.

CPE is therefore a **pore-forming enterotoxin**.

Clinical pattern

- Onset usually within approximately **6–24 hours**, often around 8–16 hours.
- Abdominal cramps.
- Diarrhoea.
- Fever and vomiting are not usually dominant.
- Illness is commonly self-limiting, but severe outcomes can occur.

C. Beta toxin is not CPE

Beta toxin is associated particularly with type C disease and necrotic enteritis.

Feature	CPE	Beta toxin
Classical association	Ordinary food poisoning	Type C necrotic enteritis
Gene/product	Enterotoxin	Distinct toxin
Disease mechanism	Claudin-associated pore formation	Severe intestinal injury/necrosis
Relation to intestinal trypsin	Do not equate with beta toxin's classic susceptibility	Beta toxin is trypsin-sensitive

Reduced intestinal proteolytic activity and dietary trypsin inhibitors can contribute to susceptibility to type C necrotic enteritis.

Alpha toxin is another distinct toxin: a phospholipase C associated with tissue injury and gas-gangrene pathogenesis. It should not be substituted for CPE in explaining ordinary food poisoning.

D. Spore resistance is strain-dependent

Food-poisoning strains carrying chromosomal **cpe** can have unusually heat-resistant spores. Spore resistance is not uniform across the species.

A cooking treatment that controls vegetative pathogens does not automatically establish adequate control of these spores.

Control logic

- Prevent prolonged warm holding.
- Use a validated cooling schedule.
- Divide large food masses when appropriate.
- Maintain suitable hot holding or refrigeration.
- Reheat using a validated process, while recognising that poor handling history cannot be judged solely from final temperature.

> ***[HPSC EXAM TRAP]***
> *- CPE, beta toxin and alpha toxin are **not interchangeable**.*
> *- Ordinary food poisoning involves toxin production principally in the intestine.*
> *- "Less than 10 minutes" refers to favourable-condition generation time.*
> *- Historical type A terminology and modern type F terminology must be reconciled, not treated as contradictory without explanation.*

3.7 *Clostridium botulinum*

First-principles model: anaerobic growth plus toxin production, not merely spore presence

Botulism risk develops when a toxigenic organism can:

35. Survive the process.
36. Germinate or grow.
37. Encounter sufficiently permissive pH, water activity and temperature.
38. Produce neurotoxin.

Low-acid, anaerobically stored foods are especially important because their environment can permit this sequence.

Basic profile

- Gram-positive rods, sometimes Gram-variable in older cultures.
- Endospore-forming.
- Obligate anaerobic growth.
- A physiologically diverse assemblage rather than one uniform processing target.

Classical toxinotypes

Toxin type	Broad classical relevance
A	Human botulism; commonly proteolytic group I
B	Human botulism; can occur in proteolytic and non-proteolytic groups
C	Predominantly animal botulism
D	Predominantly animal botulism
E	Human botulism; commonly non-proteolytic, often aquatic-food association
F	Rare human botulism
G	Rare; associated historically with group IV/*C. argentinense*

Types C/D and D/C mosaics are important in animal botulism. Modern toxin diversity extends beyond a simple seven-item list.

A. Physiological groups matter for preservation

Characteristic	Group I: proteolytic	Group II: non-proteolytic
Common toxin types	A, B, F	B, E, F
Approximate minimum growth temperature	About 10°C, depending on strain and conditions	About 3°C under permissive conditions
Approximate lower growth pH	About 4.6	About 5.0
Approximate minimum (a_w)	Around 0.94	Around 0.97
Spore heat resistance	Relatively high	Lower than group I
Refrigerated-food relevance	Growth controlled by adequate refrigeration below its growth range	Important because growth can occur at ordinary refrigeration temperatures

These are approximate physiological boundaries, not standalone validated product specifications. Solute type, acidity, temperature and formulation interact.

> ***[HPSC EXAM TRAP]***
> *"Refrigeration prevents all botulinal growth" is false. Non-proteolytic group II strains are the key exception.*

B. The pH 4.6 boundary

For low-acid canned-food processing, **equilibrium pH greater than 4.6** is a crucial classification boundary.

The reasoning is:

- Proteolytic *C. botulinum* is a central spore target in low-acid canned foods.
- Adequate acidification inhibits growth and toxin formation.
- Acidified products must achieve the required condition throughout the relevant food components.

Important qualifications:

39. pH 4.6 is not a statement that spores are killed.
40. It is not a guarantee that pre-existing toxin has been destroyed.
41. Measuring only the surrounding liquid may miss the pH of a food particle.
42. Acid-tolerant organisms may remain relevant.
43. A product still requires an appropriate validated process.

C. Botulinum neurotoxin architecture

The mature active toxin contains:

- A heavy chain of approximately 100 kDa.
- A light chain of approximately 50 kDa.
- A disulphide linkage connecting them.

The heavy chain mediates receptor interaction and translocation. The light chain is a zinc-dependent endopeptidase.

Mechanistic pathway

44. Toxin reaches peripheral cholinergic nerve endings.
45. It binds and is internalised.
46. The light chain enters the cytosol.
47. A SNARE protein is cleaved.
48. Synaptic-vesicle fusion is blocked.
49. Acetylcholine release decreases.
50. Flaccid paralysis develops.

Classical SNARE targets

Toxin type	Principal classical target
A, E	SNAP-25
B, D, F, G	Synaptobrevin/VAMP
C	SNAP-25 and syntaxin

The serotypes differ in cleavage sites and other properties.

Clinical pattern

- Cranial-nerve involvement may appear early.
- Diplopia.
- Dysphagia.
- Dysarthria.
- Descending, often symmetrical flaccid paralysis.
- Respiratory compromise in severe cases.

This is not the spastic paralysis mechanism of tetanus.

D. Food-borne versus infant botulism

Feature	Food-borne botulism	Infant botulism
Hazard ingested	Preformed toxin	Spores capable of intestinal colonisation
Main toxin-production site	Food before ingestion	Infant intestine
Disease category	Intoxication	Intestinal toxemia
Classical association	Improperly processed or stored foods	Environmental spores; honey is a preventable source

Honey should not be given to infants under 12 months.

Heating honey is not a reliable household strategy for eliminating the spore hazard.

Other forms include wound botulism and rare adult intestinal toxemia; these must not be confused with classical food-borne intoxication.

E. Toxin heat sensitivity versus spore heat resistance

Botulinum neurotoxin is substantially more heat-sensitive than spores. Public-health guidance commonly cites conditions such as at least approximately **85°C for 5 minutes** for toxin inactivation, but effectiveness depends on the matrix and process.

This is **not advice to salvage suspect food**. Suspected botulism-associated food should not be tasted or experimentally reheated.

The later thermal-kinetics module will distinguish:

[
\text{Toxin inactivation}
\neq
\text{Vegetative-cell killing}
\neq
\text{Spore destruction}
]

> ***[HPSC EXAM TRAP]***
> *- Food-borne botulism is caused by preformed toxin, not simply by swallowing spores.*
> *- Infant botulism follows spore ingestion and intestinal toxin production.*
> *- Absence of an offensive odour does not establish safety.*
> *- pH 4.6 inhibits relevant growth; it is not a universal spore-killing temperature or toxin-destruction condition.*

3.8 *Listeria monocytogenes*

First-principles model: refrigeration creates a competitive advantage

Refrigeration suppresses many organisms more strongly than it suppresses *L. monocytogenes*. Consequently, a small post-process contamination can become important during prolonged chilled storage.

The organism combines:

- Cold-growth capability.
- Environmental persistence.
- Ability to infect susceptible hosts.
- Intracellular survival and spread.

Basic profile

- Gram-positive short rods.
- Non-spore-forming.
- Catalase-positive.
- Oxidase-negative.
- Facultatively anaerobic.
- Characteristic tumbling motility, particularly evident around 20–25°C.
- Umbrella-like growth in suitable semi-solid motility media.

Psychrotrophic growth

Under suitable conditions, growth can occur near **0°C** and at ordinary refrigeration temperatures. Some experimental observations extend slightly below 0°C.

Its optimum is much warmer, commonly around **30–37°C**.

Thus:

[
\text{Ability to grow in the cold}
\neq
\text{Cold-temperature optimum}
]

Growth boundaries around pH 4.4 and (a_w) 0.92 are often used as approximate physiological screening values under otherwise favourable conditions. They must not be applied as independent, universally sufficient product specifications.

Frozen storage generally prevents multiplication but does not reliably eliminate the organism.

A. Intracellular pathogenesis

Stepwise pathway

51. Internalins facilitate entry into susceptible host cells.
52. The bacterium enters a membrane-bound vacuole.
53. Listeriolysin O and phospholipases facilitate escape.
54. Bacteria multiply in the cytoplasm.
55. ActA promotes host-actin polymerisation.
56. Actin-driven movement permits spread into adjacent cells.

[
\text{Entry}

\rightarrow
\text{vacuolar escape}
\rightarrow
\text{cytoplasmic growth}
\rightarrow
\text{actin-based movement}
\rightarrow
\text{cell-to-cell spread}
]

Listeriolysin O

- A cholesterol-dependent pore-forming cytolysin.
- Important in escape from the phagocytic vacuole.
- Its activity and regulation help balance intracellular escape with host-cell preservation.

ActA

ActA is not a digestive toxin. It is involved in recruiting host machinery for actin-based motility.

B. Disease patterns

Non-invasive illness

Febrile gastroenteritis can follow ingestion of a sufficiently contaminated food.

Invasive listeriosis

Especially important in:

- Pregnant individuals and fetuses.
- Neonates.
- Older adults.
- Immunocompromised individuals.

Manifestations include:

- Septicaemia.
- Meningitis.
- Pregnancy loss.
- Neonatal infection.

Incubation varies greatly with disease form. A single fixed incubation value is misleading.

C. Environmental persistence and biofilms

Potential harbourage sites include:

- Drains.
- Equipment joints.
- Slicers.
- Conveyor structures.
- Damaged surfaces.
- Difficult-to-clean niches.

Persistence may reflect:

- Repeated reintroduction.
- Inadequate access of cleaning agents.
- Biofilm-associated protection.
- Stress adaptation.
- Strain-level differences.

“Biofilm resilience” does not mean absolute resistance to correctly applied sanitation.

Control logic

- Prevent post-lethality contamination.
- Separate raw and ready-to-eat areas.
- Design equipment for cleaning.
- Use environmental monitoring.
- Verify sanitation effectiveness.
- Control chilled shelf life.
- Validate formulation and growth-inhibition claims.

> ***[HPSC EXAM TRAP]***
> *- Listeria is psychrotrophic, not an obligate psychrophile.*
> *- It does not form endospores.*
> *- Refrigeration is not a lethal treatment.*
> *- Listeriolysin O enables intracellular escape; ActA drives actin-based movement.*

3.9 *Salmonella enterica*

First-principles model: low water activity suppresses growth but can protect survival

A dry food may appear microbiologically stable while retaining viable *Salmonella*. Reduced water availability can increase persistence and alter heat resistance.

Therefore:

[
\text{No growth in dry storage}
\not\Rightarrow
\text{No infection risk}
]

Basic profile

- Gram-negative rods.
- Facultatively anaerobic.
- Oxidase-negative.
- Usually catalase-positive.
- Usually motile.
- Usually non-lactose-fermenting in conventional identification.
- Many produce H_2S, but exceptions occur.
- Non-spore-forming.

Non-motile serovars such as Gallinarum and Pullorum invalidate an “all salmonellae are motile” statement.

Taxonomic precision

Enteritidis, Typhimurium and Typhi are serovars, not three equivalent separate species in current usage.

A full designation can be written as:

[
\textit{Salmonella enterica}\ \text{serovar Typhimurium}
]

A. Infection pathway

57. Ingestion of viable cells.
58. Survival through gastric conditions.
59. Interaction with intestinal epithelium.
60. Type III secretion-associated invasion.
61. Inflammatory response.
62. Depending on serovar and host, local disease or systemic spread.

SPI-1-associated functions are important in invasion. SPI-2-associated functions support intracellular survival.

Disease distinction

Feature	Non-typhoidal salmonellosis	Enteric fever
Common agents	Numerous serovars, including Enteritidis and Typhimurium	Typhi and Paratyphi
Typical disease	Gastroenteritis; invasive infection possible	Systemic febrile disease
Reservoir emphasis	Often zoonotic	Predominantly human for typhoidal serovars
Food-safety classification	Infection	Infection

Incubation and infectious dose

Non-typhoidal illness can begin from approximately **6 hours to several days**, with many cases within 12–72 hours.

There is no universally valid infectious-dose number.

Dose is influenced by:

- Strain.
- Host susceptibility.
- Food buffering.
- Fat content.
- Gastric acidity.
- Physiological condition of the organism.

Some outbreaks have involved low estimated doses.

B. Heat resistance in low-(a_w) foods

Examples of relevant matrices include:

- Chocolate.
- Peanut products.
- Dry ingredients.
- Spices.
- Low-moisture powders.

Mechanistic considerations include:

- Reduced molecular mobility.
- Changes in cell physiology.
- Protective matrix effects.
- Differences in local water availability.
- Heating uniformity.

The same nominal temperature–time treatment may produce different lethality in a liquid and a dry food.

Control logic

- Prevent contamination after a kill step.
- Manage hygienic zoning in dry facilities.
- Avoid uncontrolled moisture introduction.
- Validate the lethal process in the actual formulation.
- Do not transfer liquid-food D-values directly to low-moisture foods.

> ***[HPSC EXAM TRAP]***
> *- Salmonella does not form endospores.*
> *- Desiccation tolerance does not mean growth at very low (a_w).*
> *- Low-(a_w) foods can transmit infection.*
> *- Typhi and Typhimurium are not interchangeable disease models.*

3.10 *Staphylococcus aureus*

First-principles model: human contamination followed by toxin accumulation

Food handlers can introduce *S. aureus* into a food. If conditions permit growth, enterotoxin may accumulate before the food is eaten.

The defining hazard is often:

[
\text{Preformed enterotoxin}
]

rather than viable cells present at consumption.

Basic profile

- Gram-positive cocci, commonly in clusters.
- Non-spore-forming.
- Catalase-positive.
- Usually coagulase-positive.
- Facultatively anaerobic.
- Relatively salt-tolerant.
- Able to grow at lower water activity than many other bacterial pathogens.

Growth may occur near (a_w) 0.83 under particularly favourable conditions. Enterotoxin production generally requires a higher (a_w), often around 0.86 or above under aerobic conditions, with substantial variation.

These figures do not establish a universal product safety limit.

A. Enterotoxins

Classical enterotoxins include:

- SEA.

- SEB.
- SEC.
- SED.
- SEE.

Modern recognised enterotoxin and enterotoxin-like diversity is much broader than A–E.

Two properties must be separated

Superantigen activity

Staphylococcal enterotoxins can cross-link:

- MHC class II molecules.
- Particular T-cell receptor regions.

This produces abnormal T-cell activation outside normal antigen-specific recognition.

Emetic activity

The vomiting response involves gastrointestinal and neural signalling, including vagal pathways. It should not be reduced to “superantigen activity alone explains vomiting.”

B. Thermal distinction

Target	Typical behaviour
Vegetative *S. aureus* cells	More readily destroyed by adequate cooking/pasteurisation
Preformed enterotoxin	Relatively heat-stable; may survive treatments that kill cells

“Heat-stable” is relative, not a statement of indestructibility under every possible condition.

Clinical pattern

- Onset commonly approximately **0.5–8 hours**.
- Abrupt nausea and vomiting.
- Abdominal cramps.
- Diarrhoea may occur.
- Fever is usually not prominent.

Vehicles

- Handled ready-to-eat foods.
- Cream-filled bakery products.
- Meat products.
- Salads.
- Dairy foods.

Control logic

63. Exclude contamination from infected lesions and poor handling hygiene.
64. Reduce opportunities for growth.
65. Maintain appropriate temperature control.
66. Prevent toxin accumulation before final processing.
67. Do not use absence of viable cells as proof of absence of enterotoxin.

> ***[HPSC EXAM TRAP]***
> *- S. aureus is catalase-positive; ordinary streptococci are typically catalase-negative.*
> *- It is not an endospore-former.*
> *- Coagulase does not itself explain food poisoning.*
> *- Enterotoxin A–E is a classical list, not the complete modern repertoire.*
> *- Salt curing is not automatically sufficient to prevent staphylococcal hazards.*

3.11 *Escherichia coli*: pathotypes and mechanism

First-principles model: the species name does not determine the virulence programme

Most *E. coli* are not diarrhoeagenic pathogens. Pathotypes are defined by combinations of virulence factors, including:

- Adhesins.
- Invasion systems.
- Toxins.
- Secretion systems.
- Mobile genetic elements.

A strain may combine features usually associated with different pathotypes.

Basic profile

- Gram-negative rods.
- Facultatively anaerobic.
- Oxidase-negative.
- Usually catalase-positive.
- Often motile.
- Usually lactose-fermenting, but exceptions occur.
- Non-spore-forming.

Detection of generic *E. coli* can indicate faecal contamination or hygiene failure, but it does not identify a particular pathogenic pathotype.

A. EIEC: enteroinvasive *E. coli*

Mental model

EIEC causes disease by invading and damaging colonic epithelium.

Mechanism

68. Invasion-associated factors are expressed.
69. A plasmid-associated type III secretion system supports epithelial entry.
70. Organisms escape from the entry vacuole.
71. Intracellular multiplication occurs.
72. Actin-associated spread supports movement between cells.
73. Inflammation and tissue injury produce diarrhoeal or dysenteric illness.

EIEC often exhibits biochemical features atypical of common commensal *E. coli*, including absent motility or delayed/absent lactose fermentation. These are tendencies, not universal rules.

Clinical picture

- Watery diarrhoea or dysentery.
- Abdominal pain.
- Fever may occur.
- Invasive inflammatory disease.

B. STEC and EHEC

STEC means Shiga toxin-producing *E. coli*.

EHEC is commonly used for STEC associated with the characteristic severe clinical phenotype, including haemorrhagic colitis and haemolytic uraemic syndrome.

Usage varies; the terms should not be treated as perfectly interchangeable in every context.

O157:H7

O157 and H7 identify antigenic characteristics:

- O: somatic/LPS-associated antigen.
- H: flagellar antigen.

Many important STEC are non-O157.

C. Shiga toxin: A–B_5 architecture

Component	Principal role
A subunit	Catalytic ribosome-inactivating activity
Five B subunits	Receptor binding

For classical human-disease-associated toxins, the receptor is commonly **Gb3**, globotriaosylceramide.

Mechanism

74. B subunits bind susceptible host cells.
75. Toxin is internalised.
76. Intracellular trafficking permits access of the catalytic portion.
77. The A subunit acts as an rRNA N-glycosidase.
78. A specific adenine is removed from the sarcin–ricin loop of 28S rRNA.
79. Protein synthesis is inhibited.

[
\text{28S rRNA}
\xrightarrow{\text{Shiga toxin A activity}}
\text{depurinated 28S rRNA}
]

This differs from diphtheria toxin:

[
\text{Shiga toxin: rRNA depurination}
]

[
\text{Diphtheria toxin: eEF2 ADP-ribosylation}
]

D. Attaching-and-effacing lesions

Many EHEC strains possess the locus of enterocyte effacement.

The pathway includes:

- Type III secretion.
- Tir insertion into the host membrane.
- Intimin binding to Tir.
- Actin pedestal formation.
- Microvillus effacement.

However, **not all STEC possess this system**.

E. Clinical consequences

- Abdominal cramps.
- Bloody diarrhoea in characteristic cases.
- Haemolytic uraemic syndrome:

[
\text{Microangiopathic haemolytic anaemia}
+
\text{Thrombocytopenia}
+
\text{Acute kidney injury}
]

Stx2-associated strains are particularly important in severe disease, although risk depends on toxin subtype and other factors.

Low infectious doses, sometimes estimated in the tens to hundreds of organisms, are characteristic of important outbreak strains.

F. Laboratory traps

Typical O157:H7:

- Often fails to ferment sorbitol rapidly.
- Often lacks detectable β-glucuronidase activity.

But:

- Variants occur.
- Non-O157 STEC are not reliably excluded by these characteristics.
- A generic *E. coli* test does not establish STEC status.

G. Other major diarrhoeagenic pathotypes

Pathotype	Dominant mechanism	Examination association
ETEC	LT and/or ST enterotoxins	Watery diarrhoea
EPEC	Attaching-and-effacing lesions	Infantile diarrhoea; no defining Shiga toxin
EAEC	Aggregative adherence, biofilm-associated colonisation	Persistent diarrhoea; strain heterogeneity
DAEC	Diffuse adherence	Variable clinical association
EIEC	Invasion	Inflammatory diarrhoea/dysentery

STEC	Shiga toxin	Haemorrhagic colitis/HUS

ETEC: LT versus ST

Heat-labile toxin, LT:

$$
\text{ADP-ribosylation of }G_s\alpha
\rightarrow
\text{adenylate cyclase activation}
\rightarrow
\uparrow cAMP
\rightarrow
\text{secretion}
$$

Heat-stable toxin, STa:

$$
\text{Guanylate cyclase C activation}
\rightarrow
\uparrow cGMP
\rightarrow
\text{secretion}
$$

These must not be confused with:

- Shiga toxin's rRNA depurination.
- Diphtheria toxin's eEF2 modification.

H. Hybrid strains

The O104:H4 outbreak strain demonstrated that an organism can combine an aggregative-adherence background with Shiga-toxin production.

Pathotypes are useful models, not impermeable biological compartments.

> ***[HPSC EXAM TRAP]***
> *- EIEC is invasive; STEC disease is principally toxin-mediated after infection.*
> *- EHEC does not mean all pathogenic E. coli.*
> *- O157:H7 is not the only STEC.*
> *- Shiga toxin acts on rRNA, not directly on eEF2.*
> *- LT raises cAMP; STa raises cGMP.*

3.12 Lactic acid bacteria: pathways, taxonomy and food effects

First-principles model: fermentation must regenerate NAD^+

Glycolysis requires oxidised NAD^+. In a fermentative organism, reduced NADH must be reoxidised without depending solely on respiratory electron transport.

Reduction of pyruvate to lactate provides one solution:

$$
\text{Pyruvate}+NADH+H^+
\xrightarrow{\text{lactate dehydrogenase}}
\text{Lactate}+NAD^+
$$

LAB are grouped by metabolic and practical characteristics, not by membership in one genus.

General characteristics

- Gram-positive.
- Non-spore-forming.
- Usually catalase-negative in routine conditions.
- Rods or cocci.
- Fermentative.
- Often aerotolerant or facultatively anaerobic.
- Lactic acid is a major fermentation product.

Some species can perform respiration-like metabolism when required cofactors are available. Avoid an absolute claim that no LAB can ever respire.

A. Obligately homofermentative LAB

Principal pathway

Embden–Meyerhof–Parnas pathway.

$$
[
\text{Glucose}
\rightarrow
2\text{ pyruvate}
\rightarrow
2\text{ lactate}
]
$$

A simplified net energy statement is:

$$
[
1\text{ mol glucose}
\rightarrow
2\text{ mol lactate}
+
2\text{ mol ATP}
]
$$

Examples

- *Lactobacillus delbrueckii* subsp. *bulgaricus*.
- *Lactobacillus acidophilus*.
- Many *Lactococcus*.
- *Streptococcus thermophilus*.
- Many *Pediococcus*.

Species and strain context remains important.

B. Facultatively heterofermentative LAB

These organisms generally:

- Ferment hexoses mainly through EMP, producing predominantly lactate.
- Use phosphoketolase-associated metabolism for pentoses and certain substrates.

Examples:

- *Lactiplantibacillus plantarum*.
- *Lacticaseibacillus casei*.
- *Lacticaseibacillus rhamnosus*.

Older texts usually place these species under the former broad genus *Lactobacillus*.

For an appropriate pentose:

$$
\text{Pentose}
\rightarrow
\text{Lactate}+\text{Acetate}
$$

ATP yield depends on the complete pathway and transport assumptions; approximately two ATP per pentose is a common simplified biochemical accounting.

C. Obligately heterofermentative LAB

Principal pathway

Oxidative pentose-phosphate/phosphoketolase pathway.

A key cleavage is:

$$
\text{Xylulose-5-phosphate}
\xrightarrow{\text{phosphoketolase}}
\text{Glyceraldehyde-3-phosphate}
+
\text{Acetyl phosphate}
$$

Ethanol-forming balance

$$
\text{Glucose}
\rightarrow
\text{Lactate}+\text{Ethanol}+CO_2
$$

Typical simplified net yield:

$$
1\text{ ATP per glucose}
$$

Acetate alternative

When suitable electron acceptors permit redox balance, acetyl phosphate may be directed toward acetate:

$$
\text{Acetyl phosphate}+ADP
\rightarrow
\text{Acetate}+ATP
$$

This provides additional ATP compared with the ethanol branch.

The electron-acceptor requirement must not be omitted from the explanation.

Examples

- *Leuconostoc*.
- *Oenococcus*.
- Many *Weissella*.
- *Limosilactobacillus fermentum*.
- *Limosilactobacillus reuteri*.

D. Comparative pathway matrix

Feature	Obligately homofermentative	Facultatively heterofermentative	Obligately heterofermentative
Main hexose route	EMP	Usually EMP	Phosphoketolase-associated
Main hexose products	Predominantly lactate	Predominantly lactate under suitable conditions	Lactate, CO_2, ethanol and/or acetate
CO_2 from ordinary glucose fermentation	Usually absent	Usually absent in the classical homolactic mode	Characteristic
Pentose utilisation	Restricted in the classical model	Often possible	Often possible
Simplified ATP/glucose	2	2 in homolactic mode	1 with ethanol; potentially more with acetate formation
Food effect	Rapid acidification	Substrate-dependent	Acidification plus gas and flavour metabolites

---

E. Preservation functions

LAB can inhibit undesirable organisms through:

80. Acidification.
81. Production of organic acids.
82. Bacteriocins.
83. Competition for nutrients.
84. Hydrogen peroxide in suitable conditions.
85. Other antimicrobial metabolites.

The undissociated portion of an organic acid is particularly important in membrane-associated antimicrobial effects; the detailed acid-equilibrium treatment belongs in the chemical-preservation module.

F. Spoilage and safety limitations

LAB may also cause:

- Souring.
- Gas formation.
- Package swelling.
- Slime or ropiness.
- Discolouration.
- Biogenic-amine accumulation.

Certain enterococci are opportunistic pathogens and may carry undesirable resistance determinants. Their historical use in fermentation does not make every strain a suitable starter.

G. Taxonomic correction

The broad historical genus *Lactobacillus* was reorganised in 2020. Useful examination pairings include:

Historical name	Current name
Lactobacillus plantarum	*Lactiplantibacillus plantarum*
Lactobacillus casei	*Lacticaseibacillus casei*
Lactobacillus rhamnosus	*Lacticaseibacillus rhamnosus*
Lactobacillus fermentum	*Limosilactobacillus fermentum*
Lactobacillus reuteri	*Limosilactobacillus reuteri*

Some species, including *L. delbrueckii*, remain in *Lactobacillus*.

> ***[HPSC EXAM TRAP]***
> *- Homofermentation does not mean "no ATP."*
> *- Heterofermentative ethanol production typically gives less ATP than homolactic fermentation.*
> *- Acetate formation can improve ATP yield, but redox balance still requires explanation.*
> *- LAB do not universally sterilise food by acid production.*
> *- All former Lactobacillus species were not transferred into one new genus.*

---

4. INTEGRATED COMPARATIVE MATRICES

4.1 Core identification matrix

"Typical" is essential: atypical strains and test conditions can alter individual results.

Organism/group	Gram reaction	Endospore	Typical catalase	Typical oxidase	Oxygen relationship
Acinetobacter	–	No	+	–	Aerobic
Pseudomonas	–	No	+	Usually +	Mainly respiratory/aerobic
B. cereus	+	Yes	+	Variable/not the principal discriminator	Aerobic/facultative
B. subtilis	+	Yes	+	Variable/not the principal discriminator	Mainly aerobic
Carnobacterium	+	No	Usually –	Usually –	Fermentative; tolerates low oxygen
Corynebacterium	+	No	Usually +	Species-dependent	Species-dependent
C. perfringens	+	Yes	Usually –	–	Anaerobic
C. botulinum	+	Yes	Usually –	–	Anaerobic
L. monocytogenes	+	No	+	–	Facultative
S. enterica	–	No	Usually +	–	Facultative
S. aureus	+	No	+	Usually –	Facultative
E. coli	–	No	Usually +	–	Facultative

Conventional LAB	+	No	Usually −	Usually −	Fermentative/ aerotolerant or facultative

4.2 Toxin-mechanism matrix

Organism/toxin	Principal target or action	Critical distinction
B. cereus Hbl/Nhe/CytK	Membrane injury/pore-forming activity	Diarrhoeal syndrome
B. cereus cereulide	Ionophore; mitochondrial dysfunction	Preformed, relatively heat-stable emetic toxin
C. diphtheriae toxin	ADP-ribosylation of eEF2	A catalytic, B binding/translocation
C. perfringens CPE	Claudin-associated pore formation	Intestinal production during sporulation
C. perfringens beta toxin	Severe intestinal injury	Type C necrotic enteritis
Botulinum neurotoxin	SNARE cleavage	Blocks acetylcholine release
Listeria LLO	Cholesterol-dependent pore formation	Intracellular vacuolar escape
Staphylococcal enterotoxins	Emetic and superantigen effects	Preformed toxin in food
STEC Shiga toxin	28S rRNA depurination	HUS association
ETEC LT	(G_s\alpha) modification; increased cAMP	Watery secretory diarrhoea
ETEC STa	Guanylate cyclase C; increased cGMP	Watery secretory diarrhoea

4.3 Process-selection matrix

Preservation condition	Organisms particularly relevant	Why
Refrigerated aerobic food	Psychrotrophic pseudomonads; *Listeria*	Cold-growth capability
Refrigerated vacuum-packed meat/fish	LAB, *Carnobacterium*; non-proteolytic botulinal strains if controls fail	Low oxygen changes microbial competition
Cooked food cooled slowly	*C. perfringens*, *B. cereus*	Spores survive, then germinate and grow
High-salt food	*S. aureus* and tolerant spoilage organisms	Salt is not an absolute microbial barrier
Low-moisture food	Surviving *Salmonella* and other contaminants	Growth suppression does not ensure death
Handled ready-to-eat food	*S. aureus*, *Listeria*, enteric contaminants	Post-process contamination
Low-acid hermetically sealed food	Proteolytic *C. botulinum* as a central safety target	Anaerobiosis plus permissive acidity

5. STAGE 1 — COUNT-BASED OBJECTIVE SCREENING BANK

For every question, use the same options:

- **A. Only 1 statement is correct**
- **B. Only 2 statements are correct**
- **C. All 3 statements are correct**
- **D. None of the statements is correct**

The explanations identify both the count and the specific error.

Question 1 — Gram reaction and spores

86. A Gram-positive reaction establishes that the bacterium forms endospores.
87. Old cultures of some Gram-positive organisms can appear Gram-variable.
88. The Gram-negative outer membrane contributes to envelope differences relevant to decolourisation.

Answer: B — Only statements 2 and 3 are correct.

- **1: Incorrect.** Many Gram-positive food organisms are non-spore-formers.
- **2: Correct.** Age and envelope damage can alter staining.
- **3: Correct.** Gram staining reflects envelope structure and decolourisation behaviour.

Question 2 — Catalase and oxidase

89. Catalase produces oxygen during hydrogen-peroxide decomposition.
90. Every obligate aerobe must be oxidase-positive.
91. Oxidase-negative *Acinetobacter* can still possess respiratory metabolism.

Answer: B — Only statements 1 and 3 are correct.

- **1: Correct.** (2H_2O_2\rightarrow2H_2O+O_2).
- **2: Incorrect.** The oxidase test detects particular respiratory components, not all respiration.
- **3: Correct.** Its oxidase result does not preclude aerobic metabolism.

Question 3 — Oxygen relationships

92. Facultative anaerobes may grow both with and without oxygen.
93. Aerotolerant anaerobes must use oxygen as their terminal electron acceptor.
94. Vacuum packaging kills all anaerobic bacteria.

Answer: A — Only statement 1 is correct.

- **1: Correct.** Alternative metabolic routes support growth.
- **2: Incorrect.** Oxygen tolerance is not equivalent to oxygen-dependent respiration.
- **3: Incorrect.** Vacuum is not a sterilising treatment.

Question 4 — Endospores

95. A bacterial endospore is primarily a reproductive structure.
96. Sporulation necessarily produces multiple daughter spores from one cell.
97. Endospores multiply by binary fission while dormant.

Answer: D — None is correct.

All three statements confuse survival with reproduction. Ordinary endospore formation produces a dormant survival structure; multiplication occurs after germination and outgrowth.

Question 5 — *Acinetobacter* and *Pseudomonas*

98. *Acinetobacter* is typically oxidase-negative.
99. Most familiar food-spoilage pseudomonads possess flagellar motility.
100. Both genera form heat-resistant bacterial endospores.

Answer: B — Only statements 1 and 2 are correct.

- **1: Correct.**
- **2: Correct**, with species-level exceptions.
- **3: Incorrect.** Neither genus is an endospore-forming genus.

Question 6 — *B. cereus* toxins

101. Cereulide is a cyclic depsipeptide.
102. Nhe and Hbl are identical names for cereulide.
103. Emetic illness principally follows ingestion of preformed toxin.

Answer: B — Only statements 1 and 3 are correct.

- **1: Correct.**
- **2: Incorrect.** Nhe and Hbl are protein enterotoxin systems.
- **3: Correct.** This defines the emetic intoxication model.

Question 7 — *B. cereus* disease differentiation

104. Every *B. cereus* strain produces cereulide.
105. Diarrhoeal *B. cereus* illness and emetic illness have the same toxin chemistry.
106. Reheating that kills vegetative cells invariably destroys cereulide.

Answer: D — None is correct.

- Cereulide production is strain-specific.
- Diarrhoeal toxins are proteins; cereulide is a depsipeptide.
- Cereulide can persist after a cell-lethal reheating treatment.

Question 8 — Bread rope

107. Subtilisin is a serine protease.
108. Bacillary spores can survive a baking process in the bread crumb.
109. Rope spoilage is attributable exclusively to one species and one polysaccharide.

Answer: B — Only statements 1 and 2 are correct.

- **1: Correct.**
- **2: Correct.** The food interior has its own thermal history.
- **3: Incorrect.** Related bacilli, enzymes and polymers can participate.

Question 9 — *Carnobacterium*

110. Some strains produce anti-listerial bacteriocins.
111. Some strains can form tyramine through tyrosine decarboxylation.
112. Vacuum packaging can provide conditions compatible with their growth.

Answer: C — All 3 are correct.

The statements describe compatible capabilities, not contradictory ones. A strain may offer antimicrobial potential while also requiring assessment for undesirable metabolites.

Question 10 — Diphtheria toxin

113. Fragment A mediates the principal catalytic activity.
114. Fragment B contributes to binding and translocation.
115. The toxin inhibits protein synthesis through eEF2 modification.

Answer: C — All 3 are correct.

The key reaction is ADP-ribosylation of eEF2. Do not reverse the functions of A and B.

Question 11 — Toxin-target confusion

116. Diphtheria toxin depurinates 28S rRNA as its defining mechanism.
117. Shiga toxin primarily ADP-ribosylates eEF2.
118. Botulinum neurotoxin causes paralysis principally by activating acetylcholine release.

Answer: D — None is correct.

- Diphtheria toxin modifies eEF2.
- Shiga toxin depurinates 28S rRNA.
- Botulinum toxin blocks acetylcholine release through SNARE cleavage.

Question 12 — *C. perfringens*

119. Ordinary food poisoning is classically associated with intestinal CPE production.
120. CPE and beta toxin are the same protein.
121. Generation time can be below 10 minutes under favourable conditions.

Answer: B — Only statements 1 and 3 are correct.

- **1: Correct.**
- **2: Incorrect.** Beta toxin is distinct and important in type C necrotic enteritis.
- **3: Correct.** It is an optimum-condition result, not a universal rate.

Question 13 — *C. perfringens* classification and control

122. Many strains historically described as enterotoxigenic type A are designated type F in the expanded system.
123. Cooking that kills vegetative cells necessarily kills all spores.
124. Slow cooling of bulk meat dishes can permit extensive growth.

Answer: B — Only statements 1 and 3 are correct.

The error in statement 2 is the word **necessarily**. Spores require separate consideration.

Question 14 — Botulism

125. Food-borne botulism usually follows ingestion of preformed neurotoxin.
126. Infant botulism can follow ingestion of spores.
127. Botulinum toxin light chains cleave components of the SNARE machinery.

Answer: C — All 3 are correct.

These statements connect disease classification with molecular mechanism.

Question 15 — Botulinal growth boundaries

128. pH 4.6 means all botulinal spores are destroyed.
129. Every botulinal physiological group is unable to grow at refrigeration temperatures.
130. A food without a foul odour cannot contain botulinum toxin.

Answer: D — None is correct.

- Acidity inhibits growth; it does not prove spore destruction.
- Non-proteolytic strains can grow near 3°C.
- Sensory normality does not prove safety.

Question 16 — *Listeria*

131. *L. monocytogenes* is an endospore-forming psychrophile.
132. Listeriolysin O contributes to vacuolar escape.
133. ActA contributes to actin-based intracellular movement.

Answer: B — Only statements 2 and 3 are correct.

Statement 1 contains two errors: the organism is non-spore-forming and is better described as psychrotrophic, with a substantially warmer optimum.

Question 17 — *Salmonella* in dry foods

134. Low water activity can prevent growth while allowing survival.
135. Heat resistance can increase in low-moisture matrices.
136. All serovars are non-motile.

Answer: B — Only statements 1 and 2 are correct.

Statement 3 is the reverse of the typical pattern: most salmonellae are motile, with notable exceptions.

Question 18 — Staphylococcal poisoning

137. Classical staphylococcal food poisoning is an intoxication.
138. Cell killing and enterotoxin inactivation are equivalent endpoints.
139. *S. aureus* is typically catalase-positive.

Answer: B — Only statements 1 and 3 are correct.

Enterotoxin may survive a treatment that destroys vegetative cells.

Question 19 — *E. coli* pathotypes

140. EIEC characteristically causes disease through invasion.
141. All STEC belong to serotype O157:H7.
142. Shiga toxin has an A–B_5 architecture.

Answer: B — Only statements 1 and 3 are correct.

Non-O157 STEC are important. The B pentamer mediates binding; the A component supplies catalytic activity.

Question 20 — ETEC signalling

143. LT is associated with increased intracellular cAMP.
144. STa is associated with increased intracellular cGMP.
145. LT and STa both cause disease by SNARE cleavage.

Answer: B — Only statements 1 and 2 are correct.

SNARE cleavage belongs to botulinum neurotoxin, not ETEC LT or STa.

Question 21 — LAB pathways

146. Homolactic glucose fermentation commonly yields two ATP per glucose.
147. Classical ethanol-forming heterolactic fermentation commonly yields one ATP per glucose.
148. Acetate formation from acetyl phosphate can generate additional ATP.

Answer: C — All 3 are correct.

Statement 3 still requires a compatible redox balance. Extra ATP does not eliminate the need to reoxidise reduced cofactors.

Question 22 — LAB absolutes

149. All LAB are harmless under every circumstance.
150. All LAB produce only lactic acid from every substrate.
151. All historical *Lactobacillus* species retain the same genus name.

Answer: D — None is correct.

The statements ignore opportunistic pathogenicity, spoilage, amine production, fermentation diversity and taxonomic revision.

Question 23 — Growth versus optimum

152. A psychrotrophic organism necessarily has its optimum below 5°C.
153. An organism can grow at refrigeration temperatures while having an optimum near 30°C.
154. Survival during freezing necessarily proves multiplication during frozen storage.

Answer: A — Only statement 2 is correct.

Psychrotrophy is a low-temperature growth capability. Survival and multiplication are separate events.

Question 24 — Predictive growth calculation

155. With constant generation time (g), (N_t=N_0\,2^{t/g}).

156. A population undergoing ten generations increases 1,024-fold.
157. An optimum-condition generation time can be applied unchanged through lag, cooling and nutrient depletion.

Answer: B — Only statements 1 and 2 are correct.

Statement 3 ignores the assumptions of exponential-growth modelling.

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6. STAGE 2 — CONVENTIONAL PAPER MODEL-ANSWER ARCHITECTURE

The following are proposed answer frameworks, not a verified official marking scheme. Each answer should connect **definition → mechanism → food relevance → control conclusion**.

A. TEN-MARK ANSWER FRAMEWORKS

6.1 Explain Gram reaction, catalase and oxidase tests and their food-microbiological significance. — 10 marks

Suggested allocation

Component	Marks
First-principles introduction	1
Gram-stain mechanism	3
Catalase reaction	2
Oxidase principle	2
Comparative examples and limitations	2

Model-answer structure

Introduction: Bacterial identification uses phenotypic tests that interrogate different aspects of cell-envelope structure and metabolism. No single test establishes food-safety risk.

Gram reaction:

158. Crystal violet.
159. Iodine.
160. Differential decolourisation.
161. Counterstaining.

Draw a labelled comparison showing:

- Thick peptidoglycan in a Gram-positive envelope.
- Thin peptidoglycan plus outer membrane in a Gram-negative envelope.

Catalase:

[
2H_2O_2\rightarrow2H_2O+O_2
]

Use *Staphylococcus* versus conventional LAB as the example.

Oxidase: Explain the respiratory-chain-associated reaction. Use oxidase-negative *Acinetobacter* versus usually oxidase-positive *Pseudomonas*.

Conclusion: Results are interpreted as a panel, considering culture age, reagent timing, contamination and species-level exceptions.

Essential correction: Aerobic growth does not require a positive routine oxidase test.

6.2 Compare *Acinetobacter* and *Pseudomonas* as food-spoilage bacteria. — 10 marks

Required content

162. Oxygenated-food ecological model.
163. Comparative morphology and test matrix.
164. Motility distinction.
165. Chilled-food relevance.
166. Extracellular enzyme consequences.
167. Identification limitations.

Model opening

> *Both genera include Gram-negative, non-spore-forming organisms associated with environmental contamination and food spoilage. Their significance is especially evident where oxygen and refrigerated storage select respiratory psychrotrophic flora.*

Mandatory table

Include:

- Gram reaction.
- Cell shape.
- Oxidase.
- Catalase.
- Conventional motility.
- Oxygen relationship.
- Food relevance.

Mechanistic paragraph

Psychrotrophic pseudomonads may produce proteases and lipases during raw-milk refrigeration. A later heat treatment can destroy cells without necessarily eliminating all enzyme activity, resulting in post-processing quality deterioration.

Conclusion

Control requires hygienic handling, minimisation of storage time, appropriate temperature management and attention to enzyme formation before processing—not merely final viable counts.

6.3 Differentiate diarrhoeal and emetic *Bacillus cereus* food poisoning. — 10 marks

Suggested allocation

Component	Marks
Organism and spore-survival model	1
Comparative table	4
Toxin mechanisms	3
Control and correction	2

Required comparison

- Hbl/Nhe/CytK versus cereulide.
- Protein versus depsipeptide.
- Intestinal toxin production versus preformed food toxin.
- Diarrhoea versus vomiting.
- Later versus earlier onset.
- Relative heat sensitivity.
- Typical vehicles.

Pathways

[
\text{Cells/spores ingested}
\rightarrow
\text{intestinal toxin production}
\rightarrow
\text{diarrhoea}
]

[
\text{Cereulide formed in food}
\rightarrow
\text{ingestion}
\rightarrow
\text{vomiting}
]

Control conclusion

Spore survival makes post-cooking storage important. Reheating is not a dependable corrective action after cereulide formation.

6.4 Describe rope spoilage of bread and the role of *Bacillus subtilis*. — 10 marks

Required pathway

[
\text{Spore contamination}
\rightarrow
\text{survival during baking}
\rightarrow
\text{germination}
\rightarrow
\text{enzyme/polymer production}
\rightarrow
\text{ropiness}
]

Content checklist

- Non-reproductive nature of spores.
- Bread-crumb thermal history.
- Warm and moist storage conditions.
- Amylase and protease action.
- Extracellular polysaccharide contribution.
- Sticky/stringy crumb and off-odour.
- Related bacilli.
- Preventive controls.

Enzyme equation

[
\text{Protein}+H_2O
\xrightarrow{\text{subtilisin}}
\text{smaller peptides}
]

Identify subtilisin as a serine protease with an Asp–His–Ser catalytic triad.

Conclusion

Rope prevention combines raw-material hygiene, suitable formulation, rapid cooling and storage control. It is not adequately explained by “the oven was hot.”

6.5 Discuss the beneficial and undesirable roles of *Carnobacterium* in chilled foods. — 10 marks

Answer architecture

168. Refrigeration/vacuum ecological model.
169. Identification and principal species.
170. Spoilage potential.
171. Anti-listerial bacteriocins.
172. Tyramine formation.

173.Strain-selection conclusion.

Essential equation

$$
\text{Tyrosine}+H^+
\rightarrow
\text{Tyramine}+CO_2
$$

Comparative table

Potential benefit	Corresponding limitation
Anti-listerial activity	Must be demonstrated in the food
Cold adaptation	Also permits persistence or spoilage
Bacteriocin production	Susceptibility and resistance vary
Protective-culture use	Requires strain-level safety assessment

Conclusion

A genus-level claim of usefulness is insufficient. Protective-culture suitability must include antimicrobial performance, sensory acceptability and absence of unacceptable metabolite formation.

6.6 Explain industrial amino-acid production by *C. glutamicum* and the A–B mechanism of diphtheria toxin. — 10 marks

Suggested allocation

- Genus-level distinction: 1.
- Glutamate/lysine production: 4.
- Diphtheria-toxin pathway: 4.
- Safety conclusion: 1.

Required biosynthetic equations

$$
\alpha\text{-ketoglutarate}+NH_4^{++}NADPH
\rightarrow
L\text{-glutamate}+NADP^{++}H_2O
$$

$$
\text{Oxaloacetate}
\rightarrow
\text{Aspartate}
\rightarrow
\text{Diaminopimelate}
\rightarrow
\text{Lysine}
$$

Required toxin diagram

[
B:\text{binding/translocation}
\quad\longrightarrow\quad
A:\text{eEF2 ADP-ribosylation}
]

Mention phage-associated toxigenicity and iron-responsive regulation.

Conclusion

Industrial production uses selected strains under controlled conditions. Neither industrial usefulness nor toxin production can be extrapolated to every species or strain in the genus.

6.7 Explain why *Listeria monocytogenes* is important in refrigerated ready-to-eat foods. — 10 marks

Required components

174. Psychrotrophic growth with warmer optimum.
175. Post-process contamination.
176. Prolonged chilled shelf life.
177. Environmental harbourage.
178. Intracellular virulence.
179. Susceptible populations.
180. Controls.

Mechanistic pathway

[
\text{Internalin-mediated entry}
\rightarrow
\text{LLO-assisted vacuolar escape}
\rightarrow
\text{cytoplasmic multiplication}
\rightarrow
\text{ActA-associated spread}
]

Control conclusion

Refrigeration slows growth but does not ensure elimination. A ready-to-eat-food programme requires post-lethality protection, sanitation verification, environmental monitoring and validated shelf-life/formulation controls.

6.8 Explain the persistence and thermal resistance of *Salmonella* in low-moisture foods. — 10 marks

Opening definition

Low-moisture foods may lack sufficient available water for bacterial multiplication while still permitting long-term survival of viable pathogens.

Required reasoning

181. Separate growth from survival.
182. Identify representative vehicles.
183. Explain matrix protection and low-(a_w) effects.
184. Relate these effects to heat-process validation.
185. Describe recontamination and moisture-management problems.

Essential thermal connection

At the same treatment temperature:

[
D_{\text{dry matrix}}
\text{ may exceed }
D_{\text{aqueous matrix}}
]

This is a possible and important matrix effect, not a universal numerical ratio.

Conclusion

A treatment validated in a liquid cannot be assumed effective in a dry formulation. Product-specific validation and post-process hygienic zoning are necessary.

6.9 Describe staphylococcal food poisoning and explain why reheating may fail to ensure safety. — 10 marks

Required sequence

[
\text{Human contamination}
\rightarrow
\text{growth in food}
\rightarrow
\text{enterotoxin accumulation}
\rightarrow
\text{ingestion}
\rightarrow
\text{rapid emetic illness}
]

Include

- Gram-positive clustered cocci.
- Catalase/coagulase profile.

- Salt tolerance.
- Classical enterotoxins A–E.
- Broader modern toxin repertoire.
- Heat-stable toxin versus vegetative cells.
- Typical short incubation.
- Hygiene and temperature control.

Conclusion

A satisfactory final cell count after heating cannot establish that enterotoxin was absent before heating.

B. FIFTEEN-MARK ANSWER FRAMEWORKS

6.10 Discuss *Clostridium perfringens* food poisoning, toxin differentiation and prevention. — 15 marks

Suggested allocation

Component	Marks
Organism and food ecology	2
Growth kinetics and mathematical illustration	2
CPE pathway	4
Toxin comparison and classification correction	4
Prevention	3

1. Definition and organism

An anaerobic, endospore-forming bacterium whose classical food poisoning follows ingestion of a sufficiently large viable population and intestinal enterotoxin production.

2. Food ecology

Explain:

- Spores survive cooking.
- Bulk foods cool slowly.
- Competitors may have been reduced.
- Warm interiors permit rapid growth.

3. Growth calculation

For illustrative (g=8) minutes:

[
N_t=N_0\,2^{t/8}
]

After 80 minutes:

[
N_t=1024N_0
]

State that this assumes constant favourable conditions and active exponential growth.

4. CPE mechanism

[
\text{Intestinal sporulation}
\rightarrow
\text{CPE expression}
\rightarrow
\text{mother-cell lysis}
\rightarrow
\text{claudin binding}
\rightarrow
\text{pore formation}
\rightarrow
\text{diarrhoea}
]

5. Mandatory toxin matrix

Toxin	Principal relevance
CPE	Ordinary food poisoning
Beta toxin	Type C necrotic enteritis
Alpha toxin	Phospholipase-associated tissue injury

6. Classification correction

Explain historical type A versus modern type F and the fact that not every type A strain is enterotoxigenic.

7. Prevention and operational conclusion

- Validated cooling.
- Suitable portioning.
- Appropriate hot/cold holding.
- Hygienic handling.
- Process verification.

Do not invent a jurisdictional cooling schedule unless the applicable legal or validated standard is supplied.

6.11 Explain botulism, physiological groups, toxin action and the preservation significance of pH 4.6. — 15 marks

Suggested allocation

Component	Marks
Organism and classical toxinotypes	2
Physiological-group comparison	3
Neurotoxin mechanism	4
Disease-form distinction	2

pH and process-control conclusion	4

1. Ecological model

[
\text{Surviving spores}
+
\text{anaerobiosis}
+
\text{permissive pH/}a_w/\text{temperature}
\rightarrow
\text{growth and toxin}
]

2. Toxinotypes

List A–G with broad human/animal associations. Note mosaics and additional modern diversity without obscuring the classical classification.

3. Group comparison

At minimum:

- Proteolytic group I.
- Non-proteolytic group II.
- Approximate minimum growth temperatures.
- Relative spore heat resistance.
- Refrigerated-food significance.

4. Molecular mechanism

Draw:

[
\text{Heavy-chain binding/internalisation}
\rightarrow
\text{light-chain translocation}
\rightarrow
\text{SNARE cleavage}
\rightarrow
\text{blocked acetylcholine release}
\rightarrow
\text{flaccid paralysis}
]

Include the toxin-target table.

5. Disease comparison

- Food-borne: preformed toxin.
- Infant: spores followed by intestinal toxin production.
- Honey restriction for infants under 12 months.

6. pH conclusion

An equilibrium pH at or below the relevant acidification boundary prevents growth of the central proteolytic botulinal target under appropriate conditions; it does not prove that spores or preformed toxin have been destroyed.

7. Regulatory/processing conclusion

Low-acid canned foods require a validated scheduled process. The pH classification and a thermal process serve different functions and must both be interpreted in the context of the product.

6.12 Compare EIEC, STEC/EHEC, ETEC, EPEC and EAEC. — 15 marks

Suggested allocation

Component	Marks
Pathotype concept	1
Comparative matrix	5
EIEC and STEC mechanisms	4
LT/ST signalling distinction	2
Laboratory and control implications	3

Required matrix

Pathotype	Dominant virulence mechanism	Main disease pattern
EIEC	Invasion and intracellular spread	Inflammatory diarrhoea/dysentery
STEC	Shiga toxin	Bloody diarrhoea/HUS
ETEC	LT/ST secretion pathways	Watery diarrhoea
EPEC	Attaching-and-effacing lesions	Diarrhoea, particularly in children
EAEC	Aggregative adherence/biofilm	Variable, sometimes persistent diarrhoea

EIEC pathway

[
\text{Entry}
\rightarrow
\text{vacuolar escape}
\rightarrow
\text{intracellular multiplication}
\rightarrow
\text{intercellular spread}
]

STEC pathway

[
\text{Colonisation}

\rightarrow
\text{Shiga toxin}
\rightarrow
\text{Gb3-associated cell targeting}
\rightarrow
\text{28S rRNA depurination}
\rightarrow
\text{cell injury}
]

Signalling equations

[
LT\rightarrow G_s\alpha\rightarrow\uparrow cAMP
]

[
STa\rightarrow GC\text{-}C\rightarrow\uparrow cGMP
]

Laboratory implications

- Generic *E. coli* identification does not establish pathotype.
- O157-oriented methods can miss non-O157 STEC.
- Typical sorbitol reactions are not universal.
- Virulence-factor detection may be required.

Conclusion

Food-safety interpretation should be based on the strain's virulence programme, not merely the species name or one biochemical reaction.

6.13 Derive and compare homofermentative and heterofermentative LAB pathways. — 15 marks

Suggested allocation

Component	Marks
Redox principle	2
Homolactic derivation	3
Heterolactic derivation	4
Facultative category and examples	2
Food applications, limitations and taxonomy	4

1. Begin with NAD^+ regeneration

[
\text{Pyruvate}+NADH+H^+
\rightarrow
\text{Lactate}+NAD^+
]

Explain why continuing glycolysis requires this regeneration.

2. Homolactic route

$$
\text{Glucose}
\rightarrow
2\text{ pyruvate}
\rightarrow
2\text{ lactate}
$$

Net simplified ATP:

$$
4\text{ ATP formed}-2\text{ ATP invested}
=
2\text{ ATP}
$$

3. Heterolactic route

Explain oxidative pentose-phosphate reactions, CO_2 release and phosphoketolase cleavage.

$$
\text{Xylulose-5-P}
\rightarrow
\text{Glyceraldehyde-3-P}+\text{Acetyl-P}
$$

Classical products:

$$
\text{Lactate}+\text{Ethanol}+CO_2
$$

Net simplified ATP:

$$
1\text{ ATP per glucose}
$$

4. Acetate branch

$$
\text{Acetyl-P}+ADP
\rightarrow
\text{Acetate}+ATP
$$

Discuss electron acceptors and redox balance.

5. Facultative heterofermentation

Hexoses can follow the homolactic route, while pentoses can use phosphoketolase-associated metabolism.

6. Examples and taxonomy

Provide current names with historical names where relevant.

7. Food conclusion

LAB can preserve through acidification and antimicrobial activity but also produce gas, slime, excessive acidity or biogenic amines. Starter selection is a strain-level decision.

6.14 Compare food-borne infection, intoxication and toxicoinfection using bacterial examples. — 15 marks

Suggested allocation

Component	Marks
Definitions	3
Comparative table	4
Mechanistic pathways	4
Processing implications	4

Mandatory comparison

Criterion	Infection	Intoxication	Toxicoinfection
Material ingested	Viable pathogen	Preformed toxin	Viable toxigenic organisms
Main hazardous event	Establishment of infection	Action of ingested toxin	Toxin production after ingestion
Examples	*Salmonella*, *Listeria*, STEC	*S. aureus*, emetic *B. cereus*, food-borne botulism	*C. perfringens*, diarrhoeal *B. cereus*

Mechanistic explanations

Choose one representative for each:

- *Salmonella*: invasion and inflammatory infection.
- *S. aureus*: preformed enterotoxin.
- *C. perfringens*: intestinal sporulation and CPE production.

Processing implications

186. Infection control requires viable-pathogen reduction and prevention of recontamination.
187. Intoxication control requires prevention of toxin formation; killing cells later may be inadequate.
188. Toxicoinfection control requires managing viable cells/spores and post-process growth.
189. Incubation time supports interpretation but is not a definitive diagnostic test.

Conclusion

The distinction determines which control must act before consumption: prevention of contamination, prevention of growth, destruction of organisms, prevention of toxin formation, or a validated combination.

6.15 Explain how preservation methods select bacterial communities rather than uniformly eliminating them. — 15 marks

First-principles opening

Preservation imposes environmental stress. Organisms differ in their capacity to tolerate that stress, so a process can alter the dominant flora without making the food sterile.

Required comparative matrix

Intervention	Suppressed population	Organisms/hazards still relevant
Chilling	Many mesophilic populations	Psychrotrophs, including *Listeria*
Vacuum packaging	Many obligate aerobic spoilers	LAB, facultative anaerobes and relevant anaerobic hazards
Cooking	Many vegetative cells	Surviving spores and some preformed toxins
Drying	Multiplication of many bacteria	Surviving pathogens such as *Salmonella*
Salting	Salt-sensitive organisms	Halotolerant organisms, including *S. aureus*
Acidification	Many bacterial pathogens	Acid-tolerant survivors and other spoilage organisms

Required analytical examples

- Chilling and *Listeria*.
- Vacuum packaging and *Carnobacterium*.
- Cooking/slow cooling and *C. perfringens*.
- Dry foods and *Salmonella*.
- Reheating and staphylococcal enterotoxin.

Mathematical distinction

Growth:

[
N_t=N_0e^{\mu t}
]

When a preservation condition prevents growth, the relevant (\mu) is not positive under those conditions. Nevertheless, surviving cells may remain, and death kinetics require a separate model.

Conclusion

An effective preservation system combines validated hurdles against the target hazards. Sensory stability, absence of growth and microbiological safety are related but non-equivalent outcomes.

---

7. MODULE 1 CORRECTION AND PARAMETER AUDIT

Before answering a bacterial question, apply the following audit.

7.1 Taxonomic audit

- Is the name a species, serovar, physiological group or toxinotype?
- Is historical *Lactobacillus* terminology being used?
- Does "type A *C. perfringens*" refer to the historical scheme?
- Is the *B. cereus* group being treated incorrectly as one uniform strain?

7.2 Physiological audit

- Growth capability or optimum temperature?
- Oxygen requirement or oxidase test?
- Catalase activity or general oxygen tolerance?
- Endospore survival or vegetative-cell growth?
- Growth boundary or mere survival boundary?

7.3 Toxin audit

Question	Required check
Where was toxin produced?	Food versus intestine versus infected tissue
What is its chemical nature?	Protein versus depsipeptide
What survives heating?	Cells, spores, enzymes and toxins separately
Which component is catalytic?	A/light chain as appropriate
Which host target is affected?	eEF2, rRNA, SNARE, claudin or signalling pathway
Is toxin production universal?	Usually strain-dependent

7.4 Regulatory audit

A defensible regulatory conclusion distinguishes:

190. A scientific hazard.
191. A physiological growth boundary.
192. A validated process specification.
193. A current statutory requirement.

These are not interchangeable.

For example, approximate minimum growth temperatures or water activities in this module cannot, by themselves, authorise a commercial shelf life or replace an applicable food-safety standard.

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8. REFERENCE CONTROL FOR THIS MODULE

The supplied search result did not provide the necessary primary evidence. Appropriate references for verification include:

- **FDA, *Bad Bug Book*, second edition:** organism-specific food-borne disease chapters. Its publication date must be considered when checking newer taxonomy.
- **Rood and colleagues, 2018:** expansion of the *Clostridium perfringens* toxin-based typing scheme.
- **Zheng and colleagues, 2020:** reclassification of the historical broad genus *Lactobacillus*.
- **WHO botulism guidance:** disease forms, infant-botulism prevention and toxin-related public-health guidance.

- **ICMSF food-microbiology reference works:** growth boundaries, food ecology and process-control principles.
- **The actual OUAT FSN (Ag)-321 Lessons 18–26:** required to authenticate historical wording and page-specific corrections.
- **The applicable official HPSC notification and current FSSAI regulations:** required before asserting examination-pattern or statutory compliance details.

No generic website description in the supplied result is treated here as evidence for a bacterial mechanism or regulatory threshold.

End of Module 1.

Type CONTINUE to proceed to Module 2

MODULE 2 — FOOD-BORNE MOULDS AND MYCOTOXINS

Evidence and correction note

This is an original, technically corrected study manual aligned with the requested scope. Historical course wording should be checked against the original OUAT FSN (Ag)-321 Lessons 18–26 before being described as verbatim.

Where numerical limits are mentioned, distinguish:

194. Physiological growth boundary.
195. Toxin-production optimum.
196. Processing target.
197. Legally enforceable maximum limit.

These are not interchangeable.

ESSENTIAL CORRECTION LEDGER

Issue	Scientifically correct interpretation
Moulds grow at aw 0.80	Many ordinary food moulds may grow near aw 0.80, but xerophilic moulds may grow lower. Exact limits depend on species, strain, substrate and temperature.
Minimum growth aw equals minimum toxin-production aw	Incorrect. Toxin production commonly requires a narrower and sometimes higher aw range than growth.
All moulds are harmless spoilage organisms	Incorrect. Some produce important mycotoxins.
Visible mould is the only indication of mycotoxin risk	Incorrect. Toxins may remain after visible mould disappears.
Aflatoxin is produced only by Aspergillus flavus	Incomplete. A. flavus commonly produces B aflatoxins; A. parasiticus commonly produces B and G aflatoxins. Production is strain-dependent.
Every A. flavus strain produces aflatoxin	Incorrect. Non-toxigenic strains exist.
AFM1 is produced directly in milk by mould growth	Incorrect. AFM1 is primarily a hydroxylated hepatic metabolite of ingested AFB1 excreted into milk.
Pasteurisation destroys AFM1	Not reliably. AFM1 is relatively resistant to ordinary milk-pasteurisation conditions.
Aflatoxin toxicity results from nonspecific epoxidation	Incomplete. CYP-mediated epoxidation at the terminal 8,9 double bond is central, particularly for AFB1.
P. expansum is the only patulin producer	Incomplete. It is the principal food-associated producer in apples, but other fungi may produce patulin.
Ordinary pasteurisation completely removes patulin	Incorrect. Prevention through raw-fruit sorting is essential.
Ochratoxin A comes only from A. ochraceus	Incorrect. Important producers include selected Aspergillus and Penicillium species.
Fusarium toxins are only post-harvest toxins	Incorrect. They are commonly field-associated, although levels may change during storage.
Mycotoxin limits are universal	Incorrect. Limits are commodity-, jurisdiction- and population-specific.

> ***[HPSC EXAM TRAP]*** *Mould growth, visible mould, toxin production and toxin persistence are four different propositions.*

---

1. FIRST-PRINCIPLES MODEL: WHY MOULDS GROW AND WHY MYCOTOXINS APPEAR

1.1 Ecological niche of filamentous fungi

Filamentous fungi are favoured where oxygen is available, bacterial growth is restricted by low pH or low aw, substrates are rich in carbohydrates or lipids, and storage is warm, humid or poorly ventilated.

The principal sequence is:

Spore deposition → germination → hyphal extension → mycelial colonisation → sporulation and/or toxin production

However:

Growth does not imply toxin production in every strain.

Absence of visible growth does not imply absence of toxin.

1.2 Spoilage versus mycotoxigenesis

Mould spoilage may cause visible mycelium, pigmentation, softening, dehydration, musty odours, lipolysis, pectin degradation and structural collapse.

Mycotoxigenesis is the formation of biologically active secondary metabolites by selected fungal strains under particular environmental and nutritional conditions.

A practical model is:

Mycotoxin risk = f(toxigenic strain, aw, temperature, oxygen, substrate, time, stress and competition)

1.3 Primary and secondary metabolism

Primary metabolism supports growth, energy generation, biosynthesis and reproduction. Secondary metabolism produces compounds not essential for immediate biomass formation but potentially important in ecological competition, defence or stress responses.

Many mycotoxins are secondary metabolites. Their formation may increase during drought stress, insect damage, nutrient limitation, temperature shifts or microbial competition.

> ***[HPSC EXAM TRAP]*** *Secondary metabolite does not mean unimportant; it means not directly required for basic multiplication under ordinary conditions.*

---

2. MORPHOLOGY AND PHYSIOLOGY OF FOOD MOULDS

2.1 Hyphae and mycelium

A hypha is a tubular fungal filament. A network of hyphae forms a mycelium. Hyphae may be septate or sparsely septate depending on the group. Growth occurs mainly at the hyphal tip.

Tip growth requires membrane synthesis, cell-wall assembly, vesicle trafficking and polarised cytoskeletal transport. This enables moulds to penetrate cracks, pores and damaged plant tissue.

2.2 Asexual spores: conidia

The common asexual spores of Aspergillus and Penicillium are conidia. They are produced externally on specialised structures, assist dissemination, and often tolerate drying.

Conidia are not bacterial endospores.

Conidium → germ tube → hypha → mycelium

2.3 Sexual spores

Some food moulds produce sexual spores such as ascospores. These may be more heat-resistant than vegetative hyphae or conidia in certain species. A sexual stage is not required for spoilage or mycotoxin production.

2.4 Conidiophore architecture

Aspergillus

Typical architecture:

Foot cell → conidiophore → vesicle → phialides → conidia

Species differ in stalk morphology, vesicle size, phialide arrangement, conidial colour and sclerotium formation.

Penicillium

The conidiophore branches into a brush-like penicillus containing branches, metulae, phialides and conidia. Arrangements may be monoverticillate, biverticillate or more complex.

> ***[HPSC EXAM TRAP]*** *The brush-like conidiophore is characteristic of Penicillium. Aspergillus typically forms a vesicle-based conidial head.*

---

3. WATER ACTIVITY, OXYGEN AND MOULD GROWTH

3.1 Water activity

Water activity is:

$a_w = p / p_0$

where p is the vapour pressure of water above the food and p_0 is the vapour pressure of pure water at the same temperature.

At equilibrium:

ERH (%) ≈ 100aw

Therefore, $a_w = 0.80$ corresponds approximately to 80% equilibrium relative humidity.

Total moisture percentage is insufficient because water may be bound to proteins, salts, sugars or the food matrix. Microorganisms respond more directly to available water than to total moisture.

3.2 Approximate lower aw values

Group	Approximate lower aw for growth
Many Gram-negative bacteria	About 0.95
Many Gram-positive bacteria	About 0.90–0.92
Staphylococcus aureus	Approximately 0.83–0.86, depending on conditions
Ordinary yeasts	About 0.88
Most filamentous moulds	About 0.80
Xerophilic moulds	Approximately 0.61–0.65 under favourable conditions

These are approximate physiological values, not legal product limits.

For aflatoxigenic strains, the aw required for toxin formation may be higher than the aw required for visible growth.

> ***[HPSC EXAM TRAP]*** *Mould growth at aw 0.80 does not establish aflatoxin production at aw 0.80.*

3.3 Obligate aerobic character

Food-associated filamentous moulds are generally obligately aerobic. They are favoured by oxygenated surfaces, headspace oxygen, cracks, insect channels and inadequately sealed packages.

Vacuum packaging can suppress mould growth only if oxygen is sufficiently removed and maintained at a low level. Leaks, oxygen ingress or surface exposure can restore the aerobic niche.

Reduced oxygen may inhibit moulds but select for yeasts or anaerobic bacteria. It changes the hazard profile rather than eliminating all hazards.

---

4. ENVIRONMENTAL FACTORS CONTROLLING MYCOTOXIN PRODUCTION

4.1 Temperature

Each species has minimum, optimum and maximum growth temperatures. The optimum for toxin production may differ from the optimum for biomass formation.

> ***[HPSC EXAM TRAP]*** *Maximum biomass and maximum toxin concentration are not necessarily obtained at the same temperature.*

4.2 pH

Moulds commonly tolerate a wider pH range than many bacteria and can grow in acidic foods. Many food moulds grow in approximately pH 3–6, although species vary.

Low pH suppresses many bacterial competitors and may give moulds a competitive advantage.

4.3 Oxygen and redox potential

Fungal growth and many oxidative biosynthetic reactions require oxygen. Oxygen limitation may reduce hyphal growth, sporulation and toxin biosynthesis. Oxygen may nevertheless vary greatly within a food particle or package.

4.4 Substrate

Mycotoxin production depends on carbon source, nitrogen source, lipids, trace minerals, plant defence compounds and available precursors. A substrate that supports growth does not necessarily support the same toxin profile.

4.5 Crop stress and insect damage

Pre-harvest risk factors include drought, high temperature, insect feeding, plant disease, delayed harvest and mechanical injury. Damaged kernels permit easier fungal penetration and provide more available nutrients.

4.6 Post-harvest storage

Risk rises with inadequate drying, moisture migration, condensation, warm storage, poor aeration, insect activity, prolonged storage and mixing of old and new lots.

The central control principle is:

Rapid drying + uniform moisture + cool, clean storage + lot monitoring

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5. ASPERGILLUS FLAVUS, ASPERGILLUS PARASITICUS AND AFLATOXINS

5.1 Taxonomic and ecological position

A. flavus and A. parasiticus belong to Aspergillus section Flavi.

A. flavus commonly produces B-series aflatoxins. A. parasiticus commonly produces B- and G-series aflatoxins. Some section Flavi species also produce aflatoxins, and many strains are non-toxigenic.

Species identification by colony appearance alone is insufficient to predict toxin production.

5.2 Morphological profile

A. flavus may show yellow-green to olive-green conidial masses, globose or subglobose vesicles, one or two layers of phialides and, in some strains, sclerotia.

A. parasiticus is often associated with darker green conidial masses, commonly biseriate conidial heads and production of both B- and G-series aflatoxins.

These are tendencies, not definitive diagnostic criteria.

> ***[HPSC EXAM TRAP]*** *Conidial colour is not an aflatoxin assay.*

5.3 Aflatoxin types

- Aflatoxin B1 — AFB1.
- Aflatoxin B2 — AFB2.
- Aflatoxin G1 — AFG1.
- Aflatoxin G2 — AFG2.

The historical letters refer to fluorescence: B for blue and G for green. AFB1 is generally the most important toxicologically.

5.4 Biosynthetic pathway

A simplified pathway is:

Acetate → polyketide chain → norsolorinic acid → averantin → averufin → versicolorin → sterigmatocystin → O-methylsterigmatocystin → aflatoxin

The pathway is controlled by a biosynthetic gene cluster containing genes for polyketide synthesis, oxidation, reduction, methylation, transport and regulation.

5.5 Metabolic activation of AFB1

Aflatoxin B1 is activated principally by hepatic cytochrome P450 enzymes.

AFB1 → CYP450 → AFB1-8,9-exo-epoxide

The terminal 8,9 double bond in the terminal furan ring is central to this activation.

The reactive epoxide forms DNA adducts:

AFB1-8,9-epoxide + DNA guanine → AFB1-N7-guanine adduct

Further rearrangement can generate AFB1–formamidopyrimidine adducts. These may cause misreplication, mutations, hepatocellular injury and hepatocellular carcinoma risk.

A characteristic TP53 codon-249 mutation is strongly associated with AFB1 exposure in relevant settings.

Detoxification routes include hydroxylation, glutathione conjugation and urinary excretion of metabolites.

> ***[HPSC EXAM TRAP]*** *AFB1 is not simply directly mutagenic in its original form; metabolic activation is central to its genotoxicity.*

5.6 Aflatoxin M1 and M2

AFB1 → hepatic hydroxylation → AFM1

AFB2 metabolism is associated with AFM2.

Food-chain pathway:

Contaminated feed → animal ingestion → hepatic metabolism → AFM1 in milk → human exposure

AFM1 is a milk metabolite, not usually the result of Aspergillus growing in milk. It is relatively resistant to pasteurisation, boiling and ordinary dairy processing.

Feed control is therefore a primary milk-safety intervention.

5.7 Commodity associations

Aflatoxins may occur in groundnuts, maize, tree nuts, oilseeds, cottonseed, spices, dried fruits, cereals and animal feed.

Risk is influenced by crop variety, climate, insect damage, harvest timing, drying, storage and sorting.

5.8 Prevention and control

Pre-harvest

- Resistant cultivars.
- Appropriate irrigation.
- Insect management.
- Reduction of drought stress.
- Timely harvesting.
- Approved biological control with selected non-toxigenic A. flavus strains.

Post-harvest

- Rapid drying.
- Removal of damaged kernels.
- Optical, density or colour sorting.
- Pest control.
- Moisture monitoring.
- Aerated storage.
- Prevention of condensation.
- Lot segregation.
- Representative sampling.

Aflatoxins are comparatively stable and may survive baking, frying, roasting and ordinary cooking. Physical removal, adsorption, alkaline treatment or validated detoxification may reduce exposure, but degradation products must also be assessed.

6. OCHRATOXIN A

6.1 Principal producers

Important producers include selected strains of:

- Aspergillus ochraceus sensu lato.
- A. westerdijkiae.
- A. steynii.
- A. carbonarius.
- Some A. niger-group strains.
- Penicillium verrucosum.
- Penicillium nordicum.

Warmer environments commonly favour relevant Aspergillus species, whereas P. verrucosum is important in temperate storage.

6.2 Chemical nature

OTA is a chlorinated isocoumarin linked to L-phenylalanine.

6.3 Toxicological effects

OTA has been associated with nephrotoxicity, immunotoxicity, oxidative stress, developmental toxicity and possible carcinogenicity.

6.4 Food sources

Potential sources include cereals, coffee, cocoa, dried fruits, wine, grapes, spices, nuts and animal products indirectly exposed through contaminated feed.

6.5 Control

- Rapid drying.
- Low and uniform storage moisture.
- Prevention of temperature fluctuations.
- Insect and rodent control.
- Sorting and monitoring.
- Prevention of cross-contamination.

> ***[HPSC EXAM TRAP]*** *OTA is produced by selected Aspergillus and Penicillium species; it is not exclusively an Aspergillus toxin.*

---

7. PENICILLIUM MOULDS AND THEIR TOXINS

Penicillium species may function as starter cultures, surface-ripening moulds, spoilage organisms or mycotoxin producers. Genus-level identification does not predict toxin production.

7.1 Patulin

The principal food-associated producer is Penicillium expansum, especially in bruised or rotten apples and pears, apple pomace, apple juice, cider and fruit purees.

Other fungi, including selected Aspergillus, Byssochlamys and Paecilomyces species, may produce patulin.

Patulin is an unsaturated lactone. Its reactive structure permits interaction with sulfhydryl groups in proteins and enzymes.

Patulin + protein sulfhydryl groups → enzyme dysfunction

Potential effects include gastrointestinal irritation, oxidative stress, cellular toxicity and immunotoxicity.

Apple risk factors

- Bruising.
- Fallen fruit.
- Rotten cores.
- Inadequate sorting.
- Mixing damaged and sound fruit.
- Insufficient removal of mouldy tissue.

Patulin can diffuse beyond visible mould. Removing surface mould alone does not guarantee safety.

Processing

Patulin is not reliably eliminated by ordinary pasteurisation. Reduction may depend on pH, heat, sulphur dioxide where legally permitted, ascorbic acid, fermentation, clarification and adsorption.

A commonly encountered international benchmark for patulin in apple juice is approximately 50 micrograms/kg, with lower limits sometimes applying to products for infants and young children. The current applicable national limit must be verified.

7.2 Other Penicillium toxins

Toxin	Important associations
Ochratoxin A	P. verrucosum, P. nordicum; cereals, coffee, dried foods and meat
Citrinin	P. citrinum and others; cereals, rice and stored commodities
Patulin	P. expansum; apples and apple juice
Roquefortine C	P. roqueforti and related mould-ripened products
Mycophenolic acid	Selected Penicillium species
Cyclopiazonic acid	Selected Penicillium and Aspergillus species
PR toxin	P. roqueforti under some conditions

> ***[HPSC EXAM TRAP]*** *Presence of one Penicillium species does not establish production of every toxin listed above.*

---

8. FUSARIUM MYCOTOXINS

8.1 Field infection and storage evolution

Fusarium toxins are commonly associated with pre-harvest crop infection:

Plant infection → fungal colonisation → toxin formation in the field → harvest → persistence or redistribution during storage

The absence of visible mould after cleaning does not prove absence of a Fusarium toxin.

8.2 Trichothecenes

Major trichothecenes include deoxynivalenol, nivalenol, T-2 toxin, HT-2 toxin and diacetoxyscirpenol.

Many trichothecenes interact with the 60S ribosomal subunit and inhibit protein synthesis.

DON

DON is also called vomitoxin. It is associated with Fusarium graminearum species complex, F. culmorum, wheat, barley, maize and other cereals.

Possible effects include nausea, vomiting, feed refusal, reduced growth, intestinal injury and immune modulation.

T-2 and HT-2 toxins

These may cause protein-synthesis inhibition, gastrointestinal injury, haematological effects, immunotoxicity and mucosal injury.

8.3 Zearalenone

Important producers include Fusarium graminearum species complex, F. culmorum and F. equiseti-related species.

Zearalenone is a resorcylic acid lactone with estrogenic activity:

Zearalenone + estrogen receptor → abnormal estrogenic signalling

Effects may include altered oestrus, infertility, reproductive disturbance, vulvar swelling and hyperestrogenism, particularly in pigs.

> ***[HPSC EXAM TRAP]*** *Zearalenone is principally an estrogenic/reproductive toxin, not an aflatoxin analogue.*

8.4 Fumonisins

Important producers include Fusarium verticillioides and Fusarium proliferatum. They are strongly associated with maize.

Principal compounds include fumonisins B1, B2 and B3, with FB1 commonly the principal concern.

Fumonisins resemble sphingoid-base intermediates and inhibit ceramide synthase:

Fumonisin → decreased ceramide synthase → increased sphinganine/sphingosine → disrupted sphingolipid metabolism

Possible effects include liver and kidney injury, equine leukoencephalomalacia, porcine pulmonary oedema and developmental or carcinogenic concerns.

Milling may redistribute fumonisins among bran, germ and endosperm. Alkaline processing may reduce extractable parent fumonisins but can form hydrolysed products.

8.5 Other Fusarium metabolites

- Moniliformin.
- Beauvericin.
- Enniatins.
- Fusaric acid.
- Modified or masked mycotoxins.

Cereal lots may contain several toxins simultaneously. Co-occurrence may produce additive or interactive effects.

> ***[HPSC EXAM TRAP]*** *Compliance with a limit for one mycotoxin does not establish freedom from all other mycotoxins.*

9. COMPARATIVE MATRIX

Feature	Aflatoxins	Ochratoxin A	Patulin	Zearalenone	Fumonisins
Principal fungal groups	Aspergillus section Flavi	Selected Aspergillus and Penicillium	P. expansum and others	Fusarium	Fusarium
Main food association	Groundnuts, maize, nuts and feed	Cereals, coffee and dried foods	Apples and apple juice	Cereals and maize	Maize
Principal concern	Hepatotoxicity, genotoxicity and carcinogenicity	Nephrotoxicity	Cellular and gastrointestinal toxicity	Estrogenic/ reproductive toxicity	Sphingolipid disruption
Key molecule	AFB1	Chlorinated isocoumarin–phenylalanine	Unsaturated lactone	Resorcylic acid lactone	Sphingoid-base analogue
Field/storage association	Both	Both, depending on producer	Post-harvest fruit rot	Mainly field	Mainly field
Heat stability	Relatively high	Relatively persistent	Not reliably removed by ordinary pasteurisation	Relatively persistent	Relatively persistent
Main control	Drying, sorting and storage	Moisture and storage control	Exclusion of damaged fruit	Field control and grain management	Field control and maize management

10. MYCOTOXIN SAMPLING AND ANALYTICAL CONTROL

10.1 Heterogeneous distribution

Mycotoxins are often unevenly distributed. A small number of heavily contaminated kernels may contribute disproportionately to the lot mean.

Analytical uncertainty is:

Sampling error + subsampling error + laboratory error

10.2 Correct sampling logic

Define the lot.

Collect incremental samples across locations.

Prepare an adequate aggregate sample.

Grind and homogenise thoroughly.

Take a representative laboratory subsample.

Use a validated analytical method.

Include recovery and quality-control checks.

Testing one clean-looking kernel is not representative.

10.3 Analytical methods

- Thin-layer chromatography.
- High-performance liquid chromatography.

- LC–MS/MS.
- ELISA.
- Fluorescence screening.
- Lateral-flow screening.

Screening methods are rapid; confirmatory methods provide greater specificity and quantitative confidence. A precise result from a non-representative sample may still produce an incorrect lot decision.

> ***[HPSC EXAM TRAP]*** *Analytical sensitivity is not the same as sampling representativeness.*

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11. CONTROL STRATEGY FOR MOULDS AND MYCOTOXINS

11.1 Prevention

- Good agricultural practices.
- Resistant varieties.
- Insect management.
- Timely harvesting.
- Avoidance of crop stress.
- Sanitary harvesting equipment.

11.2 Rapid post-harvest stabilisation

- Prompt drying.
- Uniform moisture reduction.
- Removal of broken and damaged material.
- Prevention of rewetting.
- Temperature control.

11.3 Storage control

- Clean structures.
- Pest exclusion.
- Moisture monitoring.
- Ventilation.
- Lot segregation.
- First-in, first-out movement.
- Periodic inspection.

11.4 Physical removal

- Sorting.
- Cleaning.
- Density separation.
- Optical sorting.
- Removal of mouldy fruit and damaged kernels.

11.5 Detoxification

Only validated processes should be used. Assessment must include residual parent toxin, degradation products, nutritional quality, sensory quality, process uniformity, legality and recontamination control.

11.6 Animal-feed control

Feed contaminated with aflatoxin can result in AFM1 in milk. Feed control is therefore a milk-safety intervention.

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12. REGULATORY THRESHOLD CAUTION

Commonly encountered international benchmarks include:

- Patulin in apple juice: often 50 micrograms/kg.
- Lower patulin limits may apply to infant and young-child products.
- AFM1 in milk is subject to very low limits in several jurisdictions, with values such as 0.05 micrograms/kg used in some regulatory systems.
- Historical Indian and international tables may use different aflatoxin limits for different foods.
- DON, OTA, zearalenone, fumonisin and trichothecene limits are commodity-specific and jurisdiction-specific.

In an HPSC answer, write:

> *The legally applicable maximum limit must be quoted from the current FSSAI contaminant and toxin schedule for the specific food category. International Codex or EU values should not be substituted silently.*

---

13. HIGH-YIELD EXAM TRAPS

13.1 Growth versus toxin production

- A toxigenic species may not produce toxin under every condition.
- A toxin may remain after the producer is dead.
- Toxin production may have a different optimum from growth.
- Visible mould removal does not guarantee toxin removal.

13.2 Fungal name versus toxin

Incorrect absolute	Corrected form
All A. flavus produce AFB1	Some strains produce AFB1; others are non-toxigenic
All A. niger produce OTA	Only selected species or strains do
All Penicillium produce patulin	Production is species- and strain-dependent
All Fusarium produce fumonisin	Fumonisin production is associated with particular species and strains

13.3 Structural traps

- AFB1 activation: terminal 8,9 double bond.
- Diphtheria toxin A: catalytic.
- Diphtheria toxin B: binding/translocation.

- Shiga toxin A: catalytic.
- Shiga toxin B5: receptor binding.
- Patulin: unsaturated lactone.
- Zearalenone: estrogenic resorcylic acid lactone.
- Fumonisin: sphingolipid-metabolism inhibitor.

13.4 Processing traps

- Roasting does not guarantee complete aflatoxin destruction.
- Pasteurisation does not reliably eliminate AFM1.
- Ordinary juice pasteurisation does not guarantee patulin removal.
- Visible mould removal is not equivalent to toxin removal.
- Drying inhibits growth but does not necessarily destroy preformed toxin.

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14. STAGE 1 — COUNT-BASED OBJECTIVE SCREENING BANK

Options:

A. Only 1 statement is correct
B. Only 2 statements are correct
C. All 3 statements are correct
D. None of the statements is correct

Question 1 — Mould morphology

Conidia of Aspergillus are bacterial endospores.
Hyphal growth occurs principally at the growing tip.
Penicillium commonly forms brush-like conidiophores.
Answer: B — Only statements 2 and 3 are correct. Conidia are fungal asexual spores, not bacterial endospores.

Question 2 — Water activity

aw = p/p0.
At equilibrium, ERH is approximately 100aw percent.
Total moisture percentage alone completely predicts mould growth.
Answer: B — Only statements 1 and 2 are correct. Water binding and availability invalidate statement 3.

Question 3 — Mould oxygen requirement

Food-associated filamentous moulds are generally obligately aerobic.
Vacuum packaging guarantees elimination of mycotoxin risk.
Oxygen availability can affect both fungal growth and secondary metabolism.
Answer: B — Only statements 1 and 3 are correct. Vacuum packaging is a hurdle, not an absolute guarantee.

Question 4 — Growth and toxin production

Every toxigenic species produces toxin whenever it grows.
The optimum for toxin production may differ from the optimum for biomass formation.
Toxin can remain after the fungus is no longer viable.
Answer: B — Only statements 2 and 3 are correct. Toxin formation is strain- and condition-dependent.

Question 5 — Aflatoxin producers

A. flavus commonly produces B-series aflatoxins.

A. parasiticus commonly produces both B- and G-series aflatoxins.

Every A. flavus strain produces aflatoxin.

Answer: B — Only statements 1 and 2 are correct. Non-toxigenic strains exist.

Question 6 — Aflatoxin fluorescence

B aflatoxins are historically associated with blue fluorescence.

G aflatoxins are historically associated with green fluorescence.

Fluorescence colour alone proves exact aflatoxin concentration.

Answer: B — Only statements 1 and 2 are correct. Quantification requires validated analysis.

Question 7 — AFB1 activation

AFB1 can be metabolically activated by cytochrome P450 enzymes.

The reactive metabolite can form DNA adducts with guanine.

AFB1 causes genotoxicity without metabolic activation in every relevant system.

Answer: B — Only statements 1 and 2 are correct. Statement 3 is an unjustified absolute.

Question 8 — Aflatoxin M1

AFM1 is associated with hydroxylation of AFB1.

AFM1 may occur in milk after animals consume contaminated feed.

AFM1 is reliably destroyed by ordinary milk pasteurisation.

Answer: B — Only statements 1 and 2 are correct. Pasteurisation is not a dependable AFM1-control step.

Question 9 — Aflatoxin structure

The terminal 8,9 double bond is important in AFB1 bioactivation.

AFB2 is another name for AFM1.

B and G aflatoxins are structurally identical except for the letter.

Answer: A — Only statement 1 is correct. AFB2 and AFM1 differ chemically, and B/G aflatoxins are structurally distinct.

Question 10 — Ochratoxin A

OTA can be produced by selected Aspergillus species.

Penicillium verrucosum is an important OTA producer.

Every strain of A. niger produces OTA.

Answer: B — Only statements 1 and 2 are correct. OTA production is strain- and species-dependent.

Question 11 — Patulin

P. expansum is strongly associated with patulin in apples.

Patulin can occur beyond visibly mouldy fruit tissue.

Ordinary pasteurisation always removes all patulin.

Answer: B — Only statements 1 and 2 are correct. Patulin is not reliably removed by ordinary pasteurisation.

Question 12 — Zearalenone

Zearalenone has estrogenic activity.

Zearalenone is principally associated with reproductive and endocrine effects.
Zearalenone is the same compound as fumonisin B1.
Answer: B — Only statements 1 and 2 are correct. They are chemically and mechanistically distinct.

Question 13 — Fumonisins

Fumonisins are strongly associated with maize.
Fumonisins interfere with sphingolipid metabolism.
Milling always destroys fumonisins.
Answer: B — Only statements 1 and 2 are correct. Milling may redistribute rather than destroy them.

Question 14 — Trichothecenes

DON is also called vomitoxin.
Trichothecenes can inhibit protein synthesis.
T-2 toxin and HT-2 toxin are aflatoxins.
Answer: B — Only statements 1 and 2 are correct. T-2 and HT-2 are trichothecenes.

Question 15 — Field versus storage moulds

Fusarium mycotoxins are commonly associated with pre-harvest crop infection.
Aspergillus and Penicillium toxins can be associated with post-harvest storage.
The field/storage distinction is absolute and has no exceptions.
Answer: B — Only statements 1 and 2 are correct. The distinction is useful but not absolute.

Question 16 — Sampling

Mycotoxin contamination may be heterogeneous within a lot.
Grinding and homogenisation reduce subsampling variation.
Testing one clean-looking kernel is sufficient to clear the lot.
Answer: B — Only statements 1 and 2 are correct. Sampling error may dominate the final uncertainty.

Question 17 — Patulin control

Exclusion of visibly damaged apples is an important preventive measure.
Patulin control should rely only on final-product heating.
Raw-material sorting can be more effective than attempting toxin removal after formation.
Answer: B — Only statements 1 and 3 are correct. Prevention is preferable to post-formation removal.

Question 18 — Mould and toxin absolutes

No visible mould means no mycotoxin.
Visible mould proves the presence of aflatoxin.
Every mould growing on food is a human pathogen.
Answer: D — None of the statements is correct. All three confuse possible association with proof.

15. STAGE 2 — CONVENTIONAL PAPER ARCHITECTURE

15.1 Ten-mark answer: Morphology and physiology of food moulds

Include:

Definition of filamentous fungi.

Hyphae and mycelium.

Tip growth.

Conidia versus bacterial endospores.

Aspergillus vesicle-based conidial head.

Penicillium brush-like penicillus.

Obligate aerobic character.

Approximate aw relationships.

Difference between growth and toxin production.

Environmental control conclusion.

15.2 Ten-mark answer: Factors affecting mycotoxin production

Discuss toxigenic strain, temperature, aw, oxygen, pH, substrate, crop stress, insect damage, storage duration and microbial competition.

Use:

Mycotoxin risk = f(strain, aw, temperature, oxygen, pH, substrate and time)

Conclude that growth and toxin production are related but non-identical.

15.3 Ten-mark answer: Aflatoxin biosynthesis and toxicity

Required pathway:

Polyketide precursors → norsolorinic acid → versicolorin → sterigmatocystin → O-methylsterigmatocystin → AFB1/AFG1

Toxicity pathway:

AFB1 → CYP450 → AFB1-8,9-exo-epoxide → AFB1-N7-guanine → mutagenesis and carcinogenesis

Mention AFB1, AFB2, AFG1, AFG2, TP53 codon-249 association, hepatic injury, carcinogenicity and strain-specific production.

15.4 Ten-mark answer: AFM1 formation in milk

Required pathway:

AFB1-contaminated feed → animal absorption → hepatic hydroxylation → AFM1 → milk excretion

State that AFM1 is a metabolite, not normally a product of mould growing in milk. Explain its relative resistance to pasteurisation and the importance of feed control.

15.5 Fifteen-mark answer: Aflatoxins, toxicology and control

Include:

Section Flavi ecology.

A. flavus and A. parasiticus comparison.

B and G aflatoxins.

Biosynthetic pathway.

CYP450 activation.

DNA adduct formation.
Hepatocellular carcinogenesis.
AFM1 milk pathway.
Pre-harvest control.
Post-harvest drying and storage.
Sorting and representative sampling.
Current food-specific legal-limit requirement.

15.6 Fifteen-mark answer: Compare patulin, OTA, zearalenone and fumonisins

Toxin	Principal food	Biochemical target	Main concern
Patulin	Apple juice	Sulfhydryl-containing cellular proteins	Gastrointestinal and cellular toxicity
OTA	Cereals, coffee and dried foods	Protein synthesis, renal and oxidative pathways	Nephrotoxicity
Zearalenone	Cereals and maize	Estrogen receptors	Reproductive and endocrine effects
Fumonisin B1	Maize	Ceramide synthase	Sphingolipid disruption

Conclude that the relevant control depends on the toxin and food chain.

15.7 Fifteen-mark answer: Sampling and analysis

Discuss lot definition, incremental sampling, aggregate sample preparation, grinding, homogenisation, laboratory subsampling, screening, confirmation, measurement uncertainty, legal decisions, documentation and traceability.

Key principle:

Reliable result = representative sample + validated method + quality assurance

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16. MASTER RECALL TABLE

Organism/toxin	Must remember
A. flavus	Commonly B aflatoxins; strain-dependent
A. parasiticus	Commonly B and G aflatoxins
AFB1	CYP450 activation; 8,9-exo-epoxide; DNA adducts
AFM1	Hydroxylated AFB1 metabolite in milk
OTA	Selected Aspergillus and Penicillium species; nephrotoxic
P. expansum	Patulin in apples and apple juice
Patulin	Unsaturated lactone; not reliably removed by ordinary pasteurisation
DON	Vomitoxin; trichothecene; protein-synthesis inhibition
Zearalenone	Estrogenic Fusarium toxin
Fumonisin B1	Maize; ceramide synthase inhibition
Mould aw	Many moulds near 0.80; xerophiles lower
Mould oxygen	Food filamentous moulds generally require oxygen

Organism/toxin	Must remember
Core trap	Growth, toxin formation and toxin persistence are distinct

Final examination rule

When answering any mycotoxin question, write in this order:

Producer → substrate → environmental conditions → toxin chemistry → molecular target → health effect → processing stability → prevention and regulatory conclusion.

MODULE 3 — FOOD-BORNE YEASTS

Evidence and correction note

Yeast growth limits, preservative resistance and statutory additive limits vary with strain, substrate, pH, temperature and jurisdiction. Physiological boundaries below are teaching values, not automatic legal standards. Current FSSAI additive schedules must be consulted for enforceable limits.

Essential correction ledger

Historical or misleading statement	Scientifically correct interpretation
All yeasts are useful fermentation organisms	Many yeasts are valuable, but several are major spoilage organisms.
High sugar prevents all microbial growth	High sugar suppresses many bacteria but selects osmophilic yeasts and xerophilic moulds.
Salt prevents yeast growth	Halotolerant and osmotolerant yeasts can grow in heavily salted foods.
Yeasts require oxygen	Many yeasts grow anaerobically by fermentation, although oxygen may increase biomass and alter metabolism.
Zygosaccharomyces bailii is merely a sugar-tolerant yeast	It is also acid-tolerant and notorious for resistance to weak-acid preservatives.
Yeast spoilage is always visible	Gas, off-flavour, turbidity or package swelling may occur before visible growth.
Debaryomyces hansenii grows only on dairy products	It is associated with salted meats, cheeses, brines and other high-salt environments.

> ***[HPSC EXAM TRAP]*** *Yeast survival, yeast growth, yeast fermentation and yeast spoilage are separate events.*

3.1 First-principles model: why yeasts survive bacterial-control hurdles

Bacteria are often disadvantaged by:

- Low pH.
- High sugar concentration.
- High salt concentration.
- Organic acids.
- Low oxygen.
- Low water activity.

Yeasts possess several countermeasures:

Flexible membrane composition.

Efficient osmotic-stress responses.

Compatible-solute accumulation, especially glycerol.

Fermentative metabolism without oxygen.

Acid-stress adaptation.

Extracellular enzyme secretion.

Biofilm or surface-film formation.

Resistance to selected preservatives.

The food-selection sequence is therefore:

Preservation hurdle → bacterial inhibition → ecological release of yeasts → yeast growth → fermentation, gas, turbidity or off-flavour

3.2 Morphology and structure

Food yeasts are unicellular fungi, usually oval, spherical, cylindrical or elongated. Important structures include:

- Cell wall containing glucans, mannoproteins and chitin.
- Plasma membrane rich in sterols, principally ergosterol.
- Nucleus and other eukaryotic organelles.
- Vacuoles for storage and stress physiology.
- Mitochondria for respiration.
- Bud scars after asexual budding.

Reproduction

Most food yeasts multiply by multilateral or polar budding:

Mother cell → bud formation → nuclear division → bud separation

Some form pseudohyphae or true hyphae. Filamentation should not automatically be interpreted as a bacterial contamination.

Sexual forms

Some yeasts produce ascospores. These are fungal sexual spores and are distinct from bacterial endospores. Ascospore formation can increase persistence under environmental stress in some species.

3.3 Respiration and fermentation

Aerobic respiration

When oxygen is available:

Glucose → pyruvate → acetyl-CoA → TCA cycle → electron transport → CO_2 + H_2O + high ATP yield

Alcoholic fermentation

Under oxygen limitation or in strongly fermentative conditions:

Glucose → 2 pyruvate → 2 acetaldehyde + 2 CO_2 → 2 ethanol

The overall reaction is:

$C_6H_{12}O_6 \rightarrow 2\ C_2H_5OH + 2\ CO_2$ + energy

The simplified net yield is approximately 2 ATP per glucose.

Crabtree effect

Some *Saccharomyces cerevisiae* strains ferment glucose even when oxygen is present if the sugar concentration is high. This is the Crabtree effect. It is a metabolic allocation phenomenon, not a statement that oxygen is absent.

> ***[HPSC EXAM TRAP]*** *Oxygen may increase yeast biomass, but it does not necessarily suppress fermentation.*

3.4 *Saccharomyces cerevisiae*

Basic profile

- Unicellular budding yeast.
- Facultatively anaerobic.
- Fermentative.
- Generally acid-tolerant.
- Used in bread, beer, wine, spirits and industrial ethanol.

Bread fermentation

The pathway is:

Flour sugars → yeast glycolysis → ethanol + CO_2

CO_2 is retained by the gluten network and expands the dough. Ethanol and volatile metabolites contribute to aroma; much ethanol is lost during baking.

Beer and wine

Yeast converts fermentable sugars into:

- Ethanol.
- Carbon dioxide.
- Glycerol.
- Organic acids.
- Higher alcohols.
- Esters.
- Aldehydes and other aroma compounds.

The final sensory profile depends on:

- Yeast strain.
- Sugar composition.
- Nitrogen supply.
- Temperature.
- Oxygen during propagation.
- Fermentation pressure.
- pH.
- Inoculum size.

Spoilage role

S. cerevisiae may spoil:

- Fruit concentrates.
- Sweet beverages.
- Syrups.
- Sauces.
- Fermented products.
- Packaged foods containing residual fermentable sugars.

Possible effects include gas formation, alcohol formation, turbidity, sediment, package swelling and flavour changes.

3.5 *Zygosaccharomyces bailii*

Ecological significance

Z. bailii is one of the most important spoilage yeasts in acidic, sugary foods.

Typical vehicles include:

- Fruit concentrates.
- Soft drinks.
- Cordials.
- Salad dressings.
- Sauces.
- Pickles.
- Jams and syrups.
- Wine and acidic beverages.

Important physiological traits

- Acid tolerance.
- Growth at relatively low water activity.
- Tolerance to high sugar.
- Resistance to some weak-acid preservatives.
- Ability to grow in the presence of sorbate or benzoate under conditions that inhibit many other yeasts.

Some strains can transform sorbic acid to 1,3-pentadiene, producing a characteristic hydrocarbon-like off-odour. Preservative resistance is strain-dependent and may be influenced by pH and adaptation history.

Control

- Hygienic filling.
- Validated preservative concentration.
- Correct pH.
- Reduction of oxygen where appropriate.
- Hot filling or pasteurisation where suitable.
- Prevention of preservative degradation.
- Control of post-process contamination.

> ***[HPSC EXAM TRAP]*** *A preservative can be legally present at its permitted level and still fail if the target yeast is resistant or the pH is unsuitable.*

3.6 *Zygosaccharomyces rouxii*

Ecological significance

Z. rouxii is highly osmotolerant and is associated with high-sugar and high-salt foods.

Typical foods include:

- Honey.
- Jams.
- Dried fruits.
- Syrups.
- Condensed products.

- Molasses.
- Soy sauce and other concentrated sauces.
- Salted or sugar-preserved foods.

Mechanism of osmotolerance

High external solute concentration lowers water activity and creates an outward osmotic gradient. The yeast responds by:

Accumulating glycerol and other compatible solutes.

Maintaining intracellular turgor.

Modifying membrane transport.

Reducing loss of cytoplasmic water.

Repairing osmotic damage.

A simplified water-stress relationship is:

High external solute → low aw → water efflux → cell stress → glycerol accumulation → partial restoration of turgor

Control

- Lower (a_w) sufficiently and uniformly.
- Combine sugar with pH control, heat treatment and hygienic packaging.
- Prevent dilution by condensation or wash-water carryover.
- Monitor concentrated products during storage.

3.7 *Debaryomyces hansenii*

Basic significance

D. hansenii is a halotolerant yeast associated with:

- Salted meats.
- Dry-cured products.
- Cheese surfaces.
- Brines.
- High-salt fish products.
- Protein-rich foods stored under reduced water activity.

Functional traits

- Strong salt tolerance.
- Ability to grow at low water activity.
- Surface-film formation.
- Lipolytic activity.
- Proteolytic activity in some strains.
- Contribution to ripening aroma in selected controlled fermentations.

The same traits can produce spoilage when growth is uncontrolled.

3.8 Other spoilage yeasts

Genus or group	Typical food relevance
Candida	Films, gas, off-flavours and spoilage of beverages and dairy foods
Pichia	Surface films on acidic beverages and fermented products;

Genus or group	Typical food relevance
	oxidative metabolism
Rhodotorula	Pink, red or orange pigmentation; dairy and surface spoilage
Hanseniaspora	Fruit and wine fermentation; volatile acidity and spoilage in some conditions
Brettanomyces or *Dekkera*	Phenolic off-flavours and beverage spoilage
Kluyveromyces	Dairy-associated fermentation and occasional spoilage
Schizosaccharomyces	Fission yeast; beverage fermentation and spoilage potential

Film-forming yeasts

Surface yeasts may use oxygen for oxidative metabolism and form a pellicle. They can:

- Consume organic acids.
- Raise surface pH.
- Reduce preservative effectiveness.
- Permit secondary bacterial growth.
- Produce musty, yeasty or solvent-like aromas.

3.9 Yeast spoilage indicators

- Gas and package swelling.
- Fermentation odour.
- Alcohol formation.
- Turbidity.
- Flocculation or sediment.
- Surface film.
- Ropy texture.
- Pigmentation.
- Pitting and softening of fruits.
- Loss of sweetness.
- Degradation of preservatives.

3.10 Yeast control matrix

Hurdle	Principal effect	Limitation
Sugar	Lowers aw	Selects osmophilic yeasts
Salt	Lowers aw and imposes ionic stress	Selects halotolerant yeasts
Low pH	Inhibits many bacteria	Acid-tolerant yeasts remain relevant
Sorbate/benzoate	Weak-acid stress	*Z. bailii* may resist
Heat	Kills vegetative cells	Recontamination and post-process survival remain possible
Vacuum or CO_2	Reduces aerobic surface growth	Fermentative yeasts may persist
Aseptic packaging	Prevents recontamination	Requires validated sterility and seal integrity

MODULE 3 — STAGE 1 OBJECTIVE BANK

Use: A = only 1 correct; B = only 2 correct; C = all 3 correct; D = none correct.

Question 1

Yeasts are eukaryotic fungi.
All yeasts form bacterial endospores.
Many yeasts reproduce by budding.
Answer: B — Statements 1 and 3 are correct.

Question 2

Alcoholic fermentation produces ethanol and carbon dioxide.
Fermentation requires molecular oxygen as the terminal electron acceptor.
Yeasts may ferment in oxygen-limited conditions.
Answer: B — Statements 1 and 3 are correct.

Question 3

High sugar can select osmophilic yeasts.
High salt can select halotolerant yeasts.
High sugar guarantees commercial sterility.
Answer: B — Statements 1 and 2 are correct.

Question 4

Z. bailii may resist weak-acid preservatives.
Z. bailii is associated with acidic beverages and concentrates.
All *Z. bailii* strains have identical preservative resistance.
Answer: B — Statements 1 and 2 are correct.

Question 5

Z. rouxii is associated with high-sugar foods.
Glycerol accumulation can assist osmotic adaptation.
Z. rouxii requires high water activity for growth.
Answer: B — Statements 1 and 2 are correct.

Question 6

D. hansenii is associated with salted foods.
It may possess lipolytic activity.
Salt tolerance proves that it cannot spoil cured foods.
Answer: B — Statements 1 and 2 are correct.

Question 7

The Crabtree effect concerns fermentative metabolism at high glucose concentration.
Oxygen always prevents fermentation by *S. cerevisiae*.
CO_2 produced by yeast leavens bread.
Answer: B — Statements 1 and 3 are correct.

Question 8

A surface film can indicate oxidative yeast growth.
Film formation always means the food is safe from bacterial growth.
Surface yeasts may consume organic acids.
Answer: B — Statements 1 and 3 are correct.

Question 9

Yeast spoilage can occur without obvious visible colonies.
Package swelling may result from fermentation.
Absence of gas proves absence of yeast.
Answer: B — Statements 1 and 2 are correct.

Question 10

Preservatives may fail because of strain resistance.
Preservative activity depends on pH.
A legally permitted preservative level guarantees control of every yeast.
Answer: B — Statements 1 and 2 are correct.

MODULE 3 — STAGE 2 ANSWER FRAMEWORKS

Ten-mark answer: Explain the spoilage of high-sugar foods by yeasts

Define yeast spoilage and explain low-pH/high-sugar ecological selection.
Describe osmotic adaptation through glycerol and compatible solutes.
Discuss *Z. rouxii* in jams, honey, syrups and dried fruit.
Discuss fermentation, gas, alcohol, turbidity and off-flavour.
Explain why sugar inhibits bacteria but not all yeasts.
Give control measures: validated aw reduction, heat, hygienic filling, pH, preservatives and packaging.
Conclude that multiple hurdles are required.

Fifteen-mark answer: Compare *S. cerevisiae*, *Z. bailii*, *Z. rouxii* and *D. hansenii*

Include morphology, oxygen relationship, major food niche, stress tolerance, spoilage effect, useful application, major control and an examination correction. The conclusion should state that the same trait can be technologically useful in one product and a spoilage advantage in another.

MODULE 4 — MICROBIAL DESTRUCTION AND THERMAL-DEATH KINETICS

Essential correction ledger

Misleading statement	Correct interpretation
A D-value is the time required to kill all organisms	It is the time for a one-log or 90% reduction under specified conditions.

Misleading statement	Correct interpretation
A z-value is a time value	It is a temperature interval associated with a tenfold change in D-value.
F and F_0 are identical at every temperature	F is referenced to a specified reference temperature; F_0 conventionally uses 121.1°C and z = 10°C.
A 12D process guarantees absolute sterility	It is a defined process target, traditionally for a specified botulinal hazard, not absolute sterility.
All survivor curves are perfectly linear	Shoulders, tailing and non-linear kinetics occur.
A process temperature is the product cold-spot temperature	Product temperature and retort temperature may differ substantially.
Commercial sterility means no microorganism remains	It means no viable organism capable of growing under the intended storage and distribution conditions, in the relevant sense.

4.1 First-principles model: lethal treatment as a population-distribution problem

A microbial population is heterogeneous. Cells may differ in:

- Age.
- Physiological state.
- Spore maturity.
- Location in the food.
- Access to heat.
- Protective matrix exposure.
- Stress history.

A thermal process therefore produces a probability distribution of survival, not an instantaneous universal kill.

Under the ideal logarithmic model:

$N_t = N_0 \times 10^{(-t/D)}$

where N_0 is the initial population, N_t is the survivors after time (t), and (D) is the decimal reduction time.

Taking logarithms:

$\log_{10} N_t = \log_{10} N_0 - t/D$

The survivor curve is linear when log survivors are plotted against time.

4.2 D-value

Definition

The D-value is the time at a specified temperature and in a specified food matrix required to reduce the surviving population by 90%, or one decimal logarithm.

D_T = time for 1-log reduction at temperature T

Graphical derivation

Plot:

- Vertical axis: $\log_{10}$ survivors.
- Horizontal axis: treatment time.

The slope is:

$\text{Slope} = -1/D$

Therefore:

$D = -1/\text{slope}$

If 10^6 cells decrease to 10^5 cells in 2 minutes, then:

$D = 2$ minutes

Example

Initial population = 10^7 CFU.

D-value = 0.5 min.

After 2 min:

Number of reductions = 2/0.5 = 4

Survivors = $10^7 \times 10^{-4} = 10^3$ CFU

Important qualifications

D-values are not universal properties of a species. They depend on:

- Temperature.
- Food composition.
- pH.
- Water activity.
- Fat.
- Salt and sugar.
- Protein.
- Spore versus vegetative state.
- Heating method.
- Recovery medium and incubation conditions.

> ***[HPSC EXAM TRAP]*** *Always write the temperature and matrix with a D-value, for example $D_{121.1}$°C in a specified food.*

4.3 Log reductions

A one-log reduction leaves 10%:

$10^6 \rightarrow 10^5$

A two-log reduction leaves 1%:

$10^6 \rightarrow 10^4$

A 12D process theoretically gives:

$10^{12} \rightarrow 10^0$

under the idealised model. However, process design does not mean that every can begins with exactly 10^{12} target spores.

4.4 z-value

Definition

The z-value is the temperature increase required to reduce the D-value by one log cycle, or to change the destruction rate tenfold.

If:

- D at T_1 = 2 min.
- D at T_2 = 0.2 min.

- $T_2 - T_1 = 10°C$.

Then:

$z = 10°C$

Graphical derivation

Plot:

- Vertical axis: $\log_{10}D$.
- Horizontal axis: temperature.

The slope is:

$Slope = -1/z$

Therefore:

$z = -1/slope$

A small z-value indicates high temperature sensitivity. A large z-value indicates lower sensitivity.

> ***[HPSC EXAM TRAP]*** *A z-value is not the number of minutes required for a log reduction.*

4.5 F-value

The F-value is the equivalent time required at a specified reference temperature to achieve a specified lethality.

For a constant temperature:

$F_T = D_T \times$ number of decimal reductions

For a changing temperature process, lethality is integrated over time.

4.6 Reference lethality and F_0

The conventional reference for F_0 is:

- Reference temperature = 121.1°C.
- z-value = 10°C.

Instantaneous lethality at temperature T is:

$L = 10^{[(T - 121.1)/z]}$

For z = 10°C:

$F_0 = \int L\, dt$

For constant temperature:

$F_0 = t \times 10^{[(T - 121.1)/10]}$

Example

A product is held at 111.1°C for 10 min.

$L = 10^{[(111.1 - 121.1)/10]} = 10^{-1} = 0.1$

$F_0 = 10 \times 0.1 = 1$ min

The same lethality can be obtained through different time–temperature combinations, provided the reference model is valid.

4.7 12D botulinum cook

The classical low-acid canned-food concept uses:

- A reference temperature of 121.1°C.
- A z-value of 10°C.

- A commonly cited D-value for proteolytic *C. botulinum* of approximately 0.21 min under defined conditions.

Then:

$12D = 12 \times 0.21 = 2.52 \text{ min } F_0$

This is a classical derivation, not a universal scheduled process.

Meaning

The 12D concept aims to provide a theoretical 12-log reduction of the specified botulinal target under the selected reference assumptions.

Limitations

- Actual spore resistance varies.
- Food matrix changes D-values.
- Container size and geometry affect heating.
- Cold-spot temperature may lag behind retort temperature.
- Acidified and low-acid foods require different process logic.
- Commercial processes are established by process authorities and validated heat-penetration studies.

> ***[HPSC EXAM TRAP]*** *12D is not a synonym for absolute sterility and not a permission to use 2.52 min in every low-acid canned food.*

4.8 Absolute sterility versus commercial sterility

Absolute sterility

Complete absence of all viable microorganisms and spores. It is a theoretical or highly stringent laboratory concept and is difficult to demonstrate directly.

Commercial sterility

A condition in which the food is free from microorganisms capable of growing under normal non-refrigerated storage and distribution conditions and is safe and stable for its intended use.

Commercial sterility depends on:

- Product pH.
- Water activity.
- Container hermeticity.
- Process severity.
- Storage temperature.
- Target organisms.

A commercially sterile food may not be absolutely sterile in the philosophical sense.

4.9 Shoulder and tailing effects

Shoulder

A delay in measurable inactivation at the start of treatment. Possible causes include:

- Clumping.
- Repair systems.
- Protective microenvironments.
- Activation requirements.

Tailing

A residual slowly declining population after the main linear phase. Possible causes include:

- Resistant subpopulations.
- Aggregation.
- Uneven heating.
- Protective matrix regions.
- Heterogeneous physiological states.

The simple D-value model may overestimate or underestimate process performance when these effects are present.

4.10 Thermal resistance modifiers

Modifier	Typical effect
Low pH	Often increases microbial heat sensitivity
High fat	May protect cells by reducing water contact
High sugar or salt	May increase thermal resistance through reduced aw
High protein	Can protect cells or spores
Large clumps	Create protected internal regions
Anaerobic conditions	May alter heat resistance and recovery
Spore maturity	Mature spores are usually more resistant
Prior heat shock	May induce stress protection

4.11 STAGE 1 objective bank

Question 1

One D-value gives a one-log reduction.
A D-value is always independent of food matrix.
A D-value must be associated with a temperature.
Answer: B — Statements 1 and 3 are correct.

Question 2

A z-value is a temperature interval.
A z-value is obtained from the log D versus temperature relationship.
A z-value is expressed in minutes.
Answer: B — Statements 1 and 2 are correct.

Question 3

F_0 conventionally uses 121.1°C and z = 10°C.
F_0 is calculated only for a constant-temperature process.
Lethality can be integrated over a changing temperature profile.
Answer: B — Statements 1 and 3 are correct.

Question 4

Commercial sterility means absolute sterility.

A hermetic seal is relevant to commercial sterility.
Storage conditions influence the meaning of commercial sterility.
Answer: B — Statements 2 and 3 are correct.

Question 5

A 12D process is a classical botulinal-process concept.
Every low-acid food requires the same exact 12D scheduled process.
Product matrix changes heat resistance.
Answer: B — Statements 1 and 3 are correct.

Question 6

A shoulder is a delayed initial reduction in a survivor curve.
Tailing can result from resistant subpopulations.
Every survivor curve is perfectly linear.
Answer: B — Statements 1 and 2 are correct.

Question 7

D121.1°C = 0.21 min and 12D gives 2.52 min under the stated assumptions.
This calculation proves every commercial retort process is adequate.
Cold-spot temperature is important in validation.
Answer: B — Statements 1 and 3 are correct.

Question 8

A lower z-value indicates greater temperature sensitivity.
F-value and D-value are identical concepts.
F-value expresses equivalent lethality at a reference temperature.
Answer: B — Statements 1 and 3 are correct.

4.12 STAGE 2 answer frameworks

Ten-mark answer: Derive D-, z- and F-values

Define each value, derive the survivor equation, show the graphical relationships, give one numerical example and explain matrix dependence.

Fifteen-mark answer: Explain the 12D botulinum cook and commercial sterility

Include low-acid food definition, spore survival, D-value assumption, 12D calculation, z-value, F_0 integration, cold-spot validation, hermetic sealing, commercial versus absolute sterility and the correction that 12D is not a universal process schedule.

MODULE 5 — LOW-TEMPERATURE PRESERVATION

Essential correction ledger

Misleading statement	Correct interpretation
Chilling kills microorganisms	Chilling generally slows growth and extends lag; it is not a

Misleading statement	Correct interpretation
	universal lethal process.
Psychrotrophs have an optimum below 5°C	They can grow near refrigeration temperature but commonly have a warmer optimum, often around 20–30°C.
Freezing sterilises food	Freezing stops or greatly slows growth but usually does not destroy all microorganisms.
Slow freezing always forms only intracellular ice	Slow freezing commonly produces large extracellular crystals, but ice location depends on rate, cell type and matrix.
Quick freezing always produces harmless intracellular crystals	Rapid freezing generally produces smaller crystals and less structural damage, but intracellular ice can still injure cells.
Frozen food cannot spoil	Chemical, enzymatic and residual microbial changes can continue slowly during frozen storage.
Thawing is a neutral operation	Thawing can cause drip loss, recrystallisation, nutrient loss and rapid microbial growth.

5.1 First-principles model: temperature changes reaction rates and phase state

Temperature affects food preservation through two separate mechanisms:

It changes biochemical and microbial reaction rates.

It changes the physical state and availability of water.

For growth, a lower temperature generally reduces metabolic rate and increases lag time. For freezing, water changes from liquid to ice, concentrating solutes in the remaining unfrozen phase.

The central sequence is:

Chilling → slower metabolism and longer lag → extended shelf life

Freezing → ice formation and freeze concentration → growth inhibition plus possible structural damage

5.2 Chilling

Chilling usually means storage above the product freezing point and commonly near 0–5°C for many foods.

Effects on microorganisms

- Slower enzyme activity.
- Reduced membrane transport.
- Longer lag phase.
- Lower specific growth rate.
- Delayed toxin formation.
- Selection for psychrotrophic organisms.

Mathematically, if the specific growth rate falls from μ_1 to μ_2:

$N_t = N_0 e^{(\mu t)}$

A lower μ gives a smaller population after the same storage time, but does not necessarily give zero growth.

Psychrophiles and psychrotrophs

Group	Growth capability	Approximate optimum
Psychrophiles	Prefer low temperatures; grow near 0°C	Often below 15°C

Group	Growth capability	Approximate optimum
Psychrotrophs	Grow at refrigeration temperatures	Often approximately 20–30°C
Mesophiles	Prefer moderate temperatures	Often 30–40°C

The exact definitions vary by textbook and experimental method.

> ***[HPSC EXAM TRAP]*** *Psychrotrophic threshold and optimum temperature are not the same parameter.*

Chilling injury in produce

Some fruits and vegetables suffer physiological chilling injury above their freezing point. Effects may include:

- Pitting.
- Surface browning.
- Failure to ripen normally.
- Water-soaked tissue.
- Increased decay susceptibility.
- Membrane leakage.

Thus the lowest temperature is not always the best storage temperature.

5.3 Freezing-point depression

Pure water freezes at 0°C at standard pressure. Dissolved solutes lower the freezing point.

The first ice forms when the liquid phase reaches its product-specific freezing point. As ice forms, solutes remain in the unfrozen phase, raising its concentration and further lowering its freezing point.

The sequence is:

`Cooling → nucleation → ice growth → solute concentration → lower remaining-phase freezing point`

5.4 Ice nucleation and crystal growth

Nucleation

The formation of a stable microscopic ice nucleus. It may be:

- Homogeneous.
- Heterogeneous, promoted by surfaces, particles or impurities.

Crystal growth

Once nuclei form, water molecules add to the crystal lattice. The final crystal size depends on:

- Cooling rate.
- Product composition.
- Nucleation temperature.
- Solute concentration.
- Tissue structure.
- Thermal conductivity.
- Package geometry.

5.5 Slow freezing

Slow freezing generally results in:

- Fewer nuclei.
- More time for crystal growth.

- Larger ice crystals.
- Greater extracellular concentration gradients.
- More mechanical disruption of tissues.
- Greater drip loss on thawing.

Large extracellular crystals can compress cells and damage membranes. Water migrates out of cells, increasing shrinkage and solute concentration.

5.6 Quick freezing

Rapid freezing generally results in:

- Greater nucleation.
- Smaller crystals.
- More uniform crystal distribution.
- Less time for solute migration.
- Better retention of texture after thawing.

The common course statement that quick freezing produces numerous small intracellular crystals is an oversimplification. The location of ice depends on freezing rate and cell dimensions. Very rapid freezing can generate intracellular ice; such ice may be lethal to cells, although small crystals generally produce less structural damage in food tissues than large crystals.

> ***[HPSC EXAM TRAP]*** *The scientifically safe comparison is large, damaging crystals in slow freezing versus generally smaller crystals and less structural damage in rapid freezing—not an absolute extracellular-versus-intracellular rule.*

5.7 Freeze concentration

When pure water crystallises as ice, solutes remain in the unfrozen liquid. Thus the unfrozen phase becomes concentrated in:

- Sugars.
- Salts.
- Organic acids.
- Proteins.
- Minerals.
- Preservatives.

Consequences include:

- Lower water activity.
- Increased ionic strength.
- Local pH shifts.
- Protein denaturation.
- Membrane stress.
- Greater concentration of reactive solutes.

5.8 Cryohydric point

The cryohydric point is the eutectic temperature at which ice and a concentrated solute phase can coexist at equilibrium for a particular aqueous solution. Foods contain multiple solutes, so they generally have a complex range of phase transitions rather than one simple universal point.

At or below a eutectic condition, solutes may crystallise rather than remain in a concentrated liquid phase. This affects:

- Texture.
- Chemical stability.
- Solute distribution.

- Recrystallisation.
- Freeze-drying behaviour.

5.9 Freeze-thaw damage

Repeated freezing and thawing causes:

- Recrystallisation.
- Membrane rupture.
- Drip loss.
- Protein denaturation.
- Emulsion destabilisation.
- Pigment migration.
- Texture softening.
- Increased microbial opportunity during thawing.

Recrystallisation

Small crystals can melt partially and feed larger crystals during temperature fluctuation. Therefore stable low-temperature storage is essential.

5.10 Microbial behaviour during freezing

Freezing may:

- Kill a fraction of vegetative cells.
- Injure membranes.
- Select resistant subpopulations.
- Preserve surviving pathogens for long periods.
- Prevent multiplication while frozen.

Survival is enhanced by:

- Sugar.
- Fat.
- Protein.
- High initial population.
- Protective food matrices.

A thawed food can support rapid growth if held in the danger zone.

5.11 Freezing process sequence

Selection and preparation.

Washing and trimming.

Blanching where required.

Rapid heat removal.

Freezing through the zone of maximum ice-crystal formation.

Completion of freezing.

Packaging with moisture and oxygen control.

Storage at stable temperature.

Controlled thawing or direct cooking.

5.12 Packaging and freezer burn

Freezer burn is a dehydration and oxidation defect caused by moisture migration from the food to the package environment.

It is encouraged by:

- Poor package barriers.
- Headspace.
- Temperature fluctuation.
- Long storage.
- Surface exposure.

Freezer burn is mainly a quality defect, not automatically a microbiological hazard, but damaged surfaces may undergo oxidation and texture deterioration.

5.13 STAGE 1 objective bank

Question 1

Chilling commonly extends the lag phase.
Chilling kills every food-borne pathogen.
Psychrotrophs may grow during refrigeration.
Answer: B — Statements 1 and 3 are correct.

Question 2

Psychrotrophic growth capability is distinct from optimum temperature.
A psychrotroph must have an optimum below 5°C.
Listeria monocytogenes is relevant to chilled foods.
Answer: B — Statements 1 and 3 are correct.

Question 3

Slow freezing commonly permits larger ice crystals.
Rapid freezing generally produces smaller crystals.
Crystal location is determined solely by the words slow or quick.
Answer: B — Statements 1 and 2 are correct.

Question 4

Freeze concentration increases solute concentration in the unfrozen phase.
Freezing can lower microbial growth without guaranteeing pathogen destruction.
Frozen storage prevents all chemical reactions.
Answer: B — Statements 1 and 2 are correct.

Question 5

Recrystallisation can occur during temperature fluctuation.
Repeated freeze-thaw cycles can increase drip loss.
Thawing makes microbial growth impossible.
Answer: B — Statements 1 and 2 are correct.

Question 6

The cryohydric point is related to eutectic phase behaviour.
Every food has one universal cryohydric temperature.
Multiple solutes complicate food freezing behaviour.
Answer: B — Statements 1 and 3 are correct.

Question 7

Chilling may select psychrotrophic spoilage flora.

Freezing can preserve surviving pathogens.

Freezing is identical to commercial sterilisation.

Answer: B — Statements 1 and 2 are correct.

5.14 STAGE 2 answer frameworks

Ten-mark answer: Compare slow and quick freezing

Define nucleation, crystal growth and freeze concentration. Compare crystal number, size, location, cellular injury, drip loss, texture and energy/process implications. Add the correction that crystal location is not an absolute rule.

Fifteen-mark answer: Explain freezing preservation from first principles

Include freezing-point depression, nucleation, phase change, latent heat, freeze concentration, aw reduction, microbial survival, tissue damage, recrystallisation, packaging, storage stability and controlled thawing.

MODULE 6 — HIGH-TEMPERATURE PRESERVATION

Essential correction ledger

Historical or misleading statement	Correct interpretation
Blanching is a sterilisation step	Blanching mainly inactivates enzymes and reduces surface microbial load; it is not commercial sterilisation.
Any peroxidase-negative result proves all enzymes are inactive	Peroxidase is a useful indicator but does not represent every enzyme or every product.
Pasteurisation makes milk sterile	It destroys specified vegetative pathogens and reduces microbial load; it does not destroy all spores.
LTLT is 62.8°C for every food	62.8°C for 30 min is a classic milk pasteurisation condition; other products and legal standards differ.
HTST is 71.7°C for all beverages	71.7°C for 15 s is the classic milk HTST condition; product-specific schedules vary.
UHT is 93.4°C for 3 s	This is not an adequate general definition of UHT milk processing. UHT commonly uses approximately 135–150°C for a few seconds, followed by aseptic packaging.
A negative phosphatase test proves no post-pasteurisation contamination	It supports adequate pasteurisation but does not exclude later contamination.
Retorting means heating the retort air to a number	Process adequacy depends on product temperature, cold spot, heat penetration and scheduled process.

6.1 First-principles model: heat must reach the target in the target state

Thermal preservation depends on:

Heat transfer from the heating medium to the package.

Heat penetration through the food.

Thermal resistance of the target organism or enzyme.

Time spent at lethal temperatures.

Cooling and post-process protection.

The process must control both:

- Microbiological safety.
- Quality deterioration from overprocessing.

A high retort temperature does not guarantee adequate lethality at the cold spot.

6.2 Blanching

Purposes

- Inactivate quality-degrading enzymes.
- Reduce surface microbial load.
- Remove intercellular gases.
- Soften tissues for filling.
- Improve colour retention in some vegetables.
- Facilitate peeling.
- Stabilise the product before freezing or drying.

Enzyme targets

Important enzyme systems include:

- Peroxidase.
- Catalase.
- Polyphenol oxidase.
- Lipoxygenase.
- Pectin methylesterase.
- Polygalacturonase.

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Peroxidase is often used as an indicator because it is relatively heat resistant and difficult to inactivate in some vegetables. Catalase is also used in selected validation schemes.

Blanching sequence

Heating to target temperature → holding → rapid cooling or immediate subsequent processing

Overblanching risks

- Texture softening.
- Leaching of soluble nutrients.
- Colour loss.
- Flavour loss.
- Excessive energy use.

Underblanching risks

- Residual enzyme activity.
- Browning.
- Off-flavour.
- Texture deterioration during frozen storage.

> ***[HPSC EXAM TRAP]*** *A blanching test is principally an enzyme-control test, not a proof of commercial sterility.*

6.3 Pasteurisation

Pasteurisation is a relatively mild thermal treatment designed to destroy specified vegetative pathogens and reduce spoilage organisms while preserving product quality.

Milk reference schedules

LTLT or batch method

62.8°C for 30 minutes

This is historically equivalent to 145°F for 30 minutes.

HTST method

71.7°C for 15 seconds

This is historically equivalent to 161°F for 15 seconds.

These are classic milk values. Current legal standards may include additional methods such as higher-temperature short-time or ultra-pasteurisation schedules.

Pasteurisation does not destroy

- All bacterial spores.
- All heat-stable toxins.
- All post-process contaminants.
- All enzymes.

6.4 Alkaline phosphatase test

Principle

Alkaline phosphatase is naturally present in raw milk and is more heat-resistant than many milk-borne vegetative pathogens. Proper pasteurisation should inactivate it below the validated detection threshold.

A substrate is hydrolysed by residual phosphatase to produce a measurable colour or fluorescence.

Interpretation

Result	Possible interpretation
Negative	Consistent with adequate pasteurisation, assuming test validity
Positive	Inadequate heating, raw-milk contamination, phosphatase from another source or analytical interference

A negative result does not prove absence of post-pasteurisation contamination.

Factors causing interpretation errors

- Incorrect incubation temperature.
- Reagent deterioration.
- Inadequate cleaning.
- Microbial phosphatases.
- Product composition.
- Excessive dilution.
- Improper sampling.

> ***[HPSC EXAM TRAP]*** *Alkaline phosphatase is a process indicator, not a direct pathogen assay.*

6.5 UHT processing

Correct definition

Ultra-high-temperature processing applies a very high temperature for a short period, commonly approximately 135–150°C for a few seconds, followed by aseptic packaging.

A typical operational range may be approximately 135–140°C for 1–10 seconds, but the actual scheduled process is product-specific.

Correction of the 93.4°C/3-second statement

A treatment at 93.4°C for 3 seconds is not a general definition of UHT commercial processing. It may describe another high-temperature treatment or a course-material misprint, but it should not be quoted as the authentic UHT standard without qualification.

Direct systems

- Steam injection.
- Steam infusion.

Advantages:

- Very rapid heating.
- Short residence time.
- Reduced quality damage if steam quality is controlled.

Indirect systems

- Plate heat exchangers.
- Tubular heat exchangers.
- Scraped-surface exchangers for viscous products.

UHT sequence

Preheating → UHT heating → holding → flash cooling or regenerative cooling → aseptic filling

Aseptic packaging requirements

- Commercially sterile product.
- Sterilised packaging material.
- Sterile filling zone.
- Hermetic closure.
- Validated sterilisation of product-contact surfaces.
- Environmental monitoring.

Shelf stability

UHT products are shelf-stable at room temperature only when commercial sterility and package integrity are maintained. After opening, refrigeration is normally required.

6.6 Retorting and commercial canning

Product categories

- Low-acid foods: equilibrium pH greater than 4.6.

- Acidified or acid foods: equilibrium pH at or below 4.6, subject to regulatory definitions and product controls.

Low-acid foods require particular attention to proteolytic *C. botulinum* spores.

Retort sequence

Container preparation and filling.

Headspace control.

Exhausting or air removal.

Seaming or hermetic closure.

Loading into the retort.

Venting and come-up period.

Holding at the scheduled process.

Cooling under controlled conditions.

Drying and inspection.

Incubation or verification where required.

Cold spot

The cold spot is the location in the container receiving the least lethal treatment. It depends on:

- Container geometry.
- Product viscosity.
- Solid-to-liquid ratio.
- Convection or conduction heating.
- Fill weight.
- Headspace.
- Agitation.

The cold spot is not automatically the geometric centre in every product.

Heat transfer modes

Conduction-heating foods

Viscous or solid foods transfer heat mainly by conduction. The cold spot is often near the geometric centre, but this must be verified.

Convection-heating foods

Liquids circulate and transfer heat by convection. The cold spot may be displaced and changes with viscosity and particle content.

Container defects

- Flipper: one end becomes convex but returns on pressure release.
- Springer: one end remains convex after the other is depressed.
- Swell: both ends bulge due to gas or pressure.
- Leaker: loss of hermetic integrity.
- Buckled seam.
- Paneling.
- Rust and corrosion.

A swollen can is unsafe to taste or open casually.

6.7 STAGE 1 objective bank

Question 1

Blanching primarily targets enzymes.
Peroxidase is often used as a heat-resistant indicator.
Blanching guarantees commercial sterility.
Answer: B — Statements 1 and 2 are correct.

Question 2

LTLT milk pasteurisation is classically 62.8°C for 30 min.
HTST milk pasteurisation is classically 71.7°C for 15 s.
These values are universal for every food product.
Answer: B — Statements 1 and 2 are correct.

Question 3

Alkaline phosphatase is a process indicator.
A negative test excludes post-pasteurisation contamination.
Phosphatase is more heat-resistant than many milk-borne vegetative pathogens.
Answer: B — Statements 1 and 3 are correct.

Question 4

Authentic UHT processing commonly uses temperatures around 135–150°C for a few seconds.
UHT products require aseptic packaging for room-temperature stability.
93.4°C for 3 seconds is the general UHT definition.
Answer: B — Statements 1 and 2 are correct.

Question 5

The retort temperature and food cold-spot temperature are identical at all times.
Heat penetration depends on product and package geometry.
Low-acid canned foods require particular attention to botulinal spores.
Answer: B — Statements 2 and 3 are correct.

Question 6

Pasteurisation destroys all bacterial spores.
Pasteurisation reduces vegetative pathogens.
Pasteurisation prevents post-process contamination.
Answer: A — Only statement 2 is correct.

Question 7

A hermetic seal is essential to protection after retorting.
A commercially sterile product is necessarily absolutely sterile.
Cooling can influence container integrity.
Answer: B — Statements 1 and 3 are correct.

6.8 STAGE 2 answer frameworks

Ten-mark answer: Compare blanching and pasteurisation

Define both, compare purpose, target, temperature severity, organisms or enzymes affected, quality effects, validation tests and limitations.

Fifteen-mark answer: Explain UHT and commercial retorting

Include thermal-death kinetics, UHT correction, direct and indirect systems, aseptic packaging, commercial sterility, retort sequence, cold spot, heat penetration, pH 4.6, container integrity and process validation.

MODULE 7 — MOISTURE REDUCTION AND HURDLE TECHNOLOGY

Essential correction ledger

Misleading statement	Correct interpretation
Moisture content and water activity are identical	Moisture is quantity; aw expresses thermodynamic availability.
Bound water is available for microbial metabolism	Strongly bound water is less available than free or weakly associated water.
aw below 0.60 makes food absolutely sterile	It prevents multiplication of ordinary microorganisms but does not destroy all cells or spores.
Drying removes all water	Drying reduces available water; residual bound water remains.
A single hurdle is always sufficient	Hurdle technology combines factors because organisms may adapt to one hurdle.
More hurdles are always better	Excessive hurdles may damage quality; the combination must be validated.
pH 4.6 alone destroys pathogens	pH 4.6 principally restricts growth of important hazards; it does not universally kill them.

7.1 First-principles model: microbial life depends on water potential, not merely water quantity

Water participates in:

- Solute transport.
- Enzyme reactions.
- Macromolecular mobility.
- Membrane function.
- DNA and protein repair.
- Cell division.

When water availability decreases:

Lower aw → osmotic stress → water efflux → reduced enzyme mobility → extended lag → slower growth → no net growth below the organism's minimum

7.2 Bound water and free water

Free or relatively mobile water

- Supports dissolution.
- Participates in biochemical reactions.
- Can migrate during drying or freezing.
- More available to microorganisms.

Bound water

- Associated with proteins, polysaccharides, salts or surfaces.
- Has reduced mobility.
- Has lower vapour pressure.
- Is less available for microbial metabolism.

Food water exists on a continuum; the terms free and bound are useful models rather than two perfectly separate pools.

7.3 Thermodynamic definition of aw

$$aw = p/p_0$$

where p is water vapour pressure above the food and p_0 is vapour pressure above pure water at the same temperature.

Water chemical potential is related to aw:

$$\mu w = \mu w^0 + RT \ln aw$$

For aw less than 1, the chemical potential of water is lower than that of pure water.

Relative humidity relationship

At equilibrium:

$$ERH (\%) \approx 100 \times aw$$

7.4 Effect of solutes

Dissolved solutes reduce aw by binding water and lowering vapour pressure.

Important humectants include:

- Sodium chloride.
- Sucrose.
- Glucose syrup.
- Fructose.
- Glycerol.
- Sorbitol.
- Propylene glycol where permitted.

The antimicrobial effect depends on:

- Solute identity.
- Concentration.
- Ionic versus non-ionic nature.
- Food pH.
- Temperature.
- Organism.

Equal moisture content does not imply equal aw if solutes differ.

7.5 Moisture sorption isotherms

A moisture sorption isotherm plots equilibrium moisture content against aw at a fixed temperature.

Adsorption

Moisture is added to a dry product.

Desorption

Moisture is removed from a wet product.

Hysteresis

The adsorption and desorption curves do not coincide. At the same aw, a food may contain different moisture levels depending on whether it is gaining or losing water.

Causes include:

- Capillary condensation.
- Swelling.
- Structural collapse.
- Solute crystallisation.
- Glass transition.
- Irreversible matrix changes.

> ***[HPSC EXAM TRAP]*** *A sorption isotherm is temperature-specific. The curve shifts when temperature changes.*

7.6 Regions of a sorption isotherm

Monolayer region

Water is strongly associated with polar sites. Mobility is low.

Multilayer region

Additional layers form with increasing mobility.

Capillary-condensation region

More mobile water accumulates in pores and can support chemical reactions and microbial growth.

The monolayer moisture content is important for predicting stability, but it is not itself a universal microbial-growth boundary.

7.7 Approximate lower growth boundary

A practical food-microbiology teaching boundary is:

$a_w \approx 0.60$

Below approximately 0.60, multiplication of ordinary food-associated microorganisms is generally prevented. Xerophilic moulds can grow near 0.61–0.65 under favourable conditions; some highly adapted organisms may approach this range.

This does not imply:

- Immediate cell death.
- Toxin destruction.
- Spore destruction.
- Elimination of viruses or prions.

- Absence of chemical deterioration.

7.8 Intermediate Moisture Foods

Intermediate Moisture Foods commonly have:

- Moisture content often approximately 15–40%, depending on definition.
- aw approximately 0.60–0.85.
- Sufficient plasticity or softness for direct consumption.
- Shelf stability without full refrigeration when properly formulated.

Examples may include:

- Fruit bars.
- Soft dried fruits.
- Meat products.
- Confectionery.
- Bakery fillings.
- Semi-moist pet foods.

IMF preservation system

Reduced aw + pH control + preservatives + packaging + heat + hygienic manufacture

Humectant mechanism

Humectants bind water and reduce aw without necessarily removing all moisture:

Food water + humectant → lower vapour pressure → lower aw

IMF limitations

- Surface moisture migration.
- Localised high-aw pockets.
- Mould growth at the surface.
- Preservative redistribution.
- Texture changes.
- Chemical browning.

7.9 Hurdle technology

Hurdle technology applies several preservation factors, each targeting a different homeostatic system.

Major hurdles:

- Temperature.
- pH.
- aw.
- Redox potential.
- Preservatives.
- Organic acids.
- Competitive flora.
- Modified atmosphere.
- Packaging integrity.
- Heat treatment.
- Irradiation.
- High pressure where applicable.

Microbial homeostasis targets

Hurdle	Disrupted function
Low pH	Proton balance and enzyme function
Low aw	Water balance and macromolecular mobility
Heat	Proteins, membranes and nucleic acids
Preservatives	Membrane transport and intracellular pH
Low Eh	Respiratory and oxidative metabolism
Cold	Enzyme kinetics and membrane fluidity
CO_2	Membrane and enzyme effects

7.10 Synergy and sequence

Hurdles may act synergistically:

Mild heat + low pH + low aw + preservative > effect of any one hurdle alone

The order of application matters. For example:

- Acidification before heating may increase heat sensitivity.
- Salt may alter preservative partitioning.
- Low aw may increase heat resistance while reducing growth.
- Packaging may prevent recontamination but also create anaerobic conditions.

Hurdles should be validated as a system, not merely listed.

7.11 Examples

Jam

- High soluble solids.
- Low aw.
- Low pH.
- Hot filling.
- Hermetic closure.
- Possible sorbate or benzoate where permitted.

Pickle

- Acetic or lactic acid.
- Salt.
- Reduced aw.
- Spices and antimicrobial compounds.
- Container integrity.

Cured meat

- Salt.
- Nitrite/nitrate.
- Reduced aw.
- Smoking.
- Refrigeration.
- Vacuum or modified atmosphere.

Fruit beverage

- Low pH.
- Pasteurisation.
- Preservative where permitted.
- Aseptic or hygienic filling.
- Cold storage after opening.

7.12 STAGE 1 objective bank

Question 1

aw is related to vapour pressure.
Moisture percentage and aw are identical.
Bound water is generally less available for microbial reactions.
Answer: B — Statements 1 and 3 are correct.

Question 2

At equilibrium, ERH is approximately 100aw percent.
aw = 0.80 corresponds approximately to 80% ERH.
The relationship is independent of temperature and equilibrium.
Answer: B — Statements 1 and 2 are correct.

Question 3

Adsorption and desorption curves can show hysteresis.
Sorption isotherms are temperature-specific.
Hysteresis proves that water activity has no thermodynamic meaning.
Answer: B — Statements 1 and 2 are correct.

Question 4

aw below 0.60 generally prevents ordinary microbial multiplication.
aw below 0.60 destroys all spores immediately.
Chemical reactions can continue in low-aw foods.
Answer: B — Statements 1 and 3 are correct.

Question 5

IMF products rely on reduced aw and other hurdles.
Surface moisture migration can create local high-aw zones.
IMF means complete dehydration.
Answer: B — Statements 1 and 2 are correct.

Question 6

Hurdle technology targets several homeostatic systems.
More hurdles are always desirable regardless of sensory quality.
Hurdles can interact synergistically.
Answer: B — Statements 1 and 3 are correct.

Question 7

Low pH may inhibit many pathogens.

Low pH universally kills all spores.

Packaging can alter redox conditions and microbial selection.

Answer: B — Statements 1 and 3 are correct.

7.13 STAGE 2 answer frameworks

Ten-mark answer: Explain water activity and moisture sorption isotherms

Define aw, derive the vapour-pressure relation, distinguish free and bound water, explain ERH, adsorption, desorption, hysteresis and microbial-growth boundaries.

Fifteen-mark answer: Explain hurdle technology with examples

Begin with microbial homeostasis, explain each hurdle, show synergistic interactions, discuss jam/pickle/cured-meat examples, identify failure modes and conclude with validation and regulatory compliance.

MODULE 8 — CHEMICAL PRESERVATIVES AND FOOD IRRADIATION DOSIMETRY

Essential correction ledger

Historical or misleading statement	Correct interpretation
Class I preservatives are unrestricted in every food	The historical Class I/Class II framework is a teaching/legal classification; current food-category permissions and maximum levels must be checked in current regulations.
All sulphites are banned	Incorrect. Permission and maximum levels depend on food category, jurisdiction, residual sulphur dioxide and declaration requirements.
Undissociated acid is irrelevant	The undissociated fraction is central to weak-organic-acid antimicrobial activity.
Nisin is active against all bacteria	Nisin is principally active against susceptible Gram-positive organisms; Gram-negative outer membranes restrict access.
Natamycin is measured in mg/dm³ for a surface application	A surface loading is an area measurement, commonly mg/dm² where specified. Volume concentration is a different quantity.
Co-137 and Ce-137 are irradiation sources	The standard radionuclide designations are cobalt-60, Co-60, and caesium-137, Cs-137.
Irradiated food becomes radioactive	Approved irradiation does not make food radioactive.
Any irradiation dose is interchangeable among foods	Dose must be commodity-, objective-, packaging- and process-specific.
Radappertisation guarantees absolute sterility	It is a high-dose irradiation concept for commercial sterility-like stability, not philosophical absolute sterility.

8.1 First-principles model: preservatives target microbial physiology

A chemical preservative is effective only when it reaches the target organism in an active form and concentration.

Effectiveness depends on:

- pH.
- Dissociation state.
- Food composition.
- Partitioning into fat or water.
- aw.
- Temperature.
- Target organism.
- Contact time.
- Initial population.
- Preservative adaptation.

The presence of a preservative in the ingredient list does not establish microbial control.

8.2 Class I and Class II preservatives

Historical Class I preservatives

Commonly listed natural or traditional preservatives include:

- Common salt.
- Sugar.
- Dextrose or glucose syrup.
- Wood smoke.
- Spices.
- Vinegar or acetic acid.
- Honey.

Historical Class II preservatives

Commonly listed chemical preservatives include:

- Benzoic acid and benzoates.
- Sorbic acid and sorbates.
- Sulphur dioxide and sulphites.
- Nitrites and nitrates.

.

The modern legal answer must follow current food-category permissions, GMP provisions, maximum levels and labelling rules. The old binary classification should not be used as a substitute for present FSSAI schedules.

8.3 Weak organic acids

Important preservatives include:

- Benzoic acid.
- Sorbic acid.
- Propionic acid.
- Acetic acid.
- Lactic acid.

.

Their strongest direct antimicrobial effect commonly comes from the undissociated molecule.

8.3.1 Henderson–Hasselbalch relationship

For a weak acid HA:

$HA \rightleftharpoons H^+ + A^-$

$pH = pKa + \log([A^-]/[HA])$

The undissociated fraction is approximately:

$\text{Fraction HA} = 1 / [1 + 10^{(pH - pKa)}]$

As pH decreases below pKa, the undissociated fraction increases.

Approximate pKa values:

Acid	Approximate pKa
Benzoic acid	4.2
Sorbic acid	4.76
Propionic acid	4.87
Acetic acid	4.76
Lactic acid	3.86

Values vary slightly with temperature and ionic environment.

8.3.2 Mechanism

Undissociated acid crosses the microbial membrane.
Intracellular pH is higher than the external acidic environment.
The acid dissociates inside the cell.
Protons lower intracellular pH.
Anions accumulate.
ATP is consumed for proton extrusion.
Enzymes and transport systems are disrupted.
Growth is inhibited or the cell dies.
External HA → membrane penetration → intracellular dissociation → proton stress + anion accumulation → metabolic failure

8.3.3 Benzoates

Most effective in acidic foods, especially beverages, pickles, sauces and fruit products where legally permitted.

Limitations:

- Reduced activity at higher pH.
- Partitioning into food components.
- Strain adaptation.
- Possible benzene formation under particular conditions involving benzoate, ascorbate, metals and heat/light; risk management is product-specific.

8.3.4 Sorbates

Sorbic acid is particularly useful against:

- Yeasts.
- Moulds.
- Some bacteria.

Applications include:

- Cheese.
- Fruit preparations.
- Beverages.
- Bakery products.
- Sauces.

.

Some spoilage yeasts, particularly *Z. bailii*, may resist or metabolise sorbate.

8.3.5 Propionates

Propionic acid and propionates are particularly useful against moulds and rope-forming bacilli in bakery products.

Their efficacy depends on:

- pH.
- Flour composition.
- Moisture.
- Distribution uniformity.
- Product temperature.

8.4 Sulphur dioxide and sulphites

Functions

- Antimicrobial effect.
- Antioxidant action.
- Inhibition of enzymatic browning.
- Reduction of non-enzymatic browning in selected systems.
- Protection of colour and flavour.

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Common compounds include:

- Sulphur dioxide.
- Sodium metabisulphite.
- Potassium metabisulphite.
- Sodium bisulphite.

.

In aqueous foods, several sulphite species coexist. The antimicrobial activity depends on pH and the fraction of molecular sulphur dioxide.

Safety and legal correction

Sulphites can provoke adverse reactions in sensitive individuals, including asthmatic reactions. This does not mean that all sulphites are universally prohibited. Current food-category permissions, residual SO_2 limits and declaration thresholds must be checked.

Factors reducing effectiveness

- High pH shifts sulphite into less active ionic species.
- Oxidation reduces active sulphite.
- Binding to aldehydes and sugars lowers free sulphite.
- Organic matter consumes available sulphite.
- Uneven distribution causes local underprotection.

8.5 Nitrites and nitrates

Technological functions

- Cured-meat colour.
- Characteristic cured flavour.
- Control of selected spoilage organisms.
- Inhibition of *C. botulinum* growth and toxin formation.
- Antioxidant effects in some cured systems.

Nitrate-to-nitrite pathway

Nitrate → nitrite → nitric oxide → nitrosylated haem pigment

Nitric oxide binds myoglobin and contributes to the cured pink colour after heating.

Safety concerns

- Methemoglobinemia from excessive nitrite exposure.
- Formation of nitrosamines under suitable conditions.
- Need for strict formulation and process control.
- Interaction with antioxidants such as ascorbate or erythorbate in cured-meat systems.

A historical statement that nitrites are permitted up to 2000 ppm in all cured meats should not be used as a current universal legal value. Current limits are food-category-specific and may be much lower.

8.6 Biological antimicrobials

8.6.1 Nisin

Nisin is a lantibiotic peptide produced by selected strains of *Lactococcus lactis*.

Mechanism

Nisin binds lipid II.

Cell-wall precursor transport is disrupted.

Nisin–lipid II complexes form membrane pores.

Ion gradients collapse.

The susceptible Gram-positive cell dies.

Nisin + lipid II → pore formation + cell-wall inhibition → Gram-positive cell death

Spectrum

Nisin is mainly active against Gram-positive bacteria, including selected clostridia and bacilli. Gram-negative organisms are often protected by their outer membrane.

Applications

- Processed cheese.
- Dairy products.
- Selected canned or heat-processed foods.
- Control of clostridial outgrowth in permitted products.

Limitations

- Resistance or tolerance.
- Inactivation by food proteins or enzymes.
- Uneven distribution.
- Reduced activity at unsuitable pH.
- Not a substitute for validated thermal processing.

8.6.2 Natamycin

Natamycin is a polyene antifungal produced by *Streptomyces natalensis* and related organisms.

Mechanism

Natamycin binds ergosterol in fungal membranes and disrupts membrane function. It has limited penetration into many food matrices and is therefore useful primarily as a surface treatment.

Applications

- Surface treatment of cheese.
- Selected cured meats.
- Other permitted surface applications.

Unit correction

A surface limit is an area-based quantity, for example mg/dm² where specified. mg/dm³ expresses a volume concentration and should not be substituted for an area loading.

> ***[HPSC EXAM TRAP]*** *Natamycin is antifungal, whereas nisin is principally antibacterial against Gram-positive organisms.*

8.7 Food irradiation

8.7.1 Physical principle

Ionising radiation transfers sufficient energy to remove electrons or produce reactive chemical species.

Food irradiation can produce:

- Direct DNA damage.
- Water radiolysis.
- Hydroxyl radicals.
- Hydrogen atoms.
- Hydrated electrons.
- Oxidative injury to membranes and proteins.

Water radiolysis

H_2O + radiation → •OH + •H + e^-aq + H_2O_2 + molecular products

The actual product spectrum depends on oxygen, solutes, dose rate and food composition.

8.7.2 Radiation sources

Gamma radiation

- Cobalt-60, **Co-60**.

- Caesium-137, **Cs-137**.

Correction: The isotope designation is Cs-137, not Ce-137. Ce denotes cerium; the standard food-irradiation source is caesium-137.

Electron beams

- Machine-generated electrons.
- High dose rate.
- Limited penetration compared with gamma and high-energy X-rays.
- Suitable for relatively thin or surface-oriented products.

X-rays

- Machine-generated.
- Greater penetration than electron beams.
- Require conversion of electron energy into X-rays.
- Efficiency is generally lower than direct electron treatment.

8.7.3 Dosimetry units

Gray

1 Gy = 1 joule of absorbed energy per kilogram of material

Kilogray

1 kGy = 1000 Gy

Dose is absorbed energy, not exposure time alone.

Important process quantities:

- Minimum dose.
- Maximum dose.
- Average dose.
- Dose uniformity ratio.
- Dose mapping.
- Product density and package geometry.

8.7.4 Radiation D10 value

The D10 value is the radiation dose required for a one-log or 90% reduction in a specified organism under specified conditions.

D10 = dose for 1-log reduction

It is analogous conceptually to a thermal D-value but is not numerically interchangeable with it.

Radiation resistance depends on:

- Organism.
- Spore or vegetative state.
- Temperature.
- Oxygen.
- Food composition.
- Water activity.
- Dose rate.
- Recovery conditions.

8.7.5 Historical process terminology

Radurisation

Low-dose irradiation intended to extend shelf life by reducing spoilage flora, broadly analogous to mild pasteurisation.

Radicidation

Dose intended to reduce specific non-spore-forming pathogens to a defined level.

Radappertisation

High-dose treatment intended to achieve commercial sterility-like stability by reducing viable organisms, including more resistant forms, to a very low level.

These terms are historical or descriptive process terminology. Current regulations may use different legal language.

8.7.6 Indicative technical dose bands

The following are commonly taught or operational ranges, not a replacement for current Indian legal schedules:

Objective	Commonly encountered dose range	Important qualification
Sprout inhibition in potatoes, onions and garlic	Approximately 0.05–0.15 kGy	Commodity and notification specific
Insect disinfestation of grains and pulses	Approximately 0.25–1.0 kGy	Insect species and commodity matter
Quarantine insect treatment of fruits	Approximately 0.15–1.0 kGy	Specific pest and importing-country requirement matter
Delay of ripening or senescence	Approximately 0.3–1.0 kGy	Variety and physiological maturity matter
Reduction of vegetative pathogens in meat or seafood	Approximately 1–5 kGy	Product temperature and sensory quality matter
Spice and dry-herb decontamination	Approximately 5–10 kGy	Dose must comply with current schedule
High-dose commercial-sterility-like treatment	Approximately 25–45 kGy or higher in selected systems	Food quality and regulatory authorisation are critical

Statutory caution

Exact approved dose clearances in India must be taken from the current FSSAI regulations, relevant Department of Atomic Energy or AERB requirements, irradiation-plant licence conditions and commodity-specific notifications. A historical textbook band must not be presented as a current universal statutory limit.

8.7.7 Food irradiation does not make food radioactive

Approved irradiation uses energy that passes through or is absorbed by the product. It does not leave a radioactive source in the food. The distinction is between:

- Irradiation: exposure to ionising energy.
- Radioactive contamination: unwanted radioactive material deposited in the food.

These are not the same event.

8.7.8 Factors affecting irradiation effectiveness

- Dose uniformity.
- Product thickness.
- Density.
- Packaging geometry.
- Oxygen concentration.
- Temperature.
- Water activity.
- Initial microbial load.
- Spore resistance.
- Food composition.
- Post-irradiation recontamination.

Irradiation does not replace hygienic manufacture.

8.7.9 Labelling and process control

A controlled irradiation operation requires:

Approved food and purpose.
Validated dose range.
Dosimeter calibration.
Dose mapping.
Product identity and lot traceability.
Source security or accelerator control.
Packaging compatibility.
Microbiological verification.
Required labelling.
Record retention.

MODULE 8 — STAGE 1 OBJECTIVE BANK

Use: A = only 1 correct; B = only 2 correct; C = all 3 correct; D = none correct.

Question 1 — Weak acids

Undissociated weak acid can cross microbial membranes.
Lower pH generally increases the undissociated fraction.
The dissociation state is independent of pH.
Answer: B — Statements 1 and 2 are correct.

Question 2 — Preservative spectrum

Sorbates are important against yeasts and moulds.
Propionates are useful against moulds in bakery products.
Preservatives control every organism equally.
Answer: B — Statements 1 and 2 are correct.

Question 3 — Sulphites

Sulphites can act as antioxidants.
All sulphites are universally banned.
Current permissions depend on food category and jurisdiction.

Answer: B — Statements 1 and 3 are correct.

Question 4 — Nitrite

Nitrite contributes to cured-meat colour through nitric oxide chemistry.
Nitrite is relevant to control of *C. botulinum*.
Nitrite has no toxicological concerns.
Answer: B — Statements 1 and 2 are correct.

Question 5 — Nisin

Nisin binds lipid II.
Nisin is principally active against susceptible Gram-positive organisms.
Nisin is a polyene antifungal.
Answer: B — Statements 1 and 2 are correct.

Question 6 — Natamycin

Natamycin binds fungal ergosterol.
It is commonly used as a surface antifungal.
A surface loading must be expressed as a volume concentration.
Answer: B — Statements 1 and 2 are correct.

Question 7 — Irradiation sources

Co-60 is cobalt-60.
Cs-137 is caesium-137.
Ce-137 is the standard designation for caesium-137.
Answer: B — Statements 1 and 2 are correct.

Question 8 — Dosimetry

One Gy equals one joule per kilogram absorbed dose.
One kGy equals 1000 Gy.
Dose is measured solely as exposure time.
Answer: B — Statements 1 and 2 are correct.

Question 9 — Irradiation terminology

Radurisation is a low-dose shelf-life extension concept.
Radicidation concerns reduction of specified pathogens.
Radappertisation means absolute sterility in every case.
Answer: B — Statements 1 and 2 are correct.

Question 10 — Radiation safety

Approved irradiation makes food radioactive.
Radiation can damage microbial DNA directly and indirectly.
Water radiolysis can generate reactive species.
Answer: B — Statements 2 and 3 are correct.

Question 11 — Dose ranges

Sprout-inhibition doses are generally lower than spice-decontamination doses.
One dose is suitable for every commodity and purpose.

Dose uniformity must be validated.

Answer: B — Statements 1 and 3 are correct.

Question 12 — Legal limits

Current additive limits are food-category-specific.

A historical textbook value is automatically a current statutory limit.

Irradiation authorisation must be verified against current regulations.

Answer: B — Statements 1 and 3 are correct.

MODULE 8 — STAGE 2 ANSWER FRAMEWORKS

Ten-mark answer: Explain the antimicrobial mechanism of weak organic acids

Define weak-acid dissociation, give the Henderson–Hasselbalch relationship, derive the undissociated fraction, explain membrane penetration, intracellular dissociation, proton stress, anion accumulation and ATP depletion. Compare benzoate, sorbate and propionate and conclude with pH dependence.

Ten-mark answer: Compare nisin and natamycin

Include microbial source, chemical class, target membrane component, spectrum, food application, surface or bulk use, limitations and the trap that natamycin is antifungal while nisin is mainly active against Gram-positive bacteria.

Fifteen-mark answer: Discuss chemical preservatives and their safety significance

Organise the answer as:

Definition and purpose.

Historical Class I/Class II classification.

Current regulatory correction.

Weak organic acids and pH.

Sulphites.

Nitrites and nitrates.

Biological antimicrobials.

Food-category and maximum-level control.

Labelling and sensitive consumers.

Hurdle-technology conclusion.

Fifteen-mark answer: Explain food irradiation, radiation sources, dosimetry and applications

Include radiation physics, direct and indirect DNA damage, Co-60, Cs-137, electron beams, X-rays, Gy and kGy, D10 value, radurisation, radicidation, radappertisation, indicative dose bands, sprout inhibition, insect disinfestation, spice decontamination, dose mapping, statutory verification and the fact that irradiation does not make food radioactive.

FINAL CROSS-MODULE EXAMINATION AUDIT

Before writing a preservation answer, classify the target:

Vegetative cell.
Endospore.
Fungal conidium.
Toxin.
Enzyme.
Virus.
Quality attribute.

Then classify the process:

Growth inhibition.
Cell killing.
Spore destruction.
Toxin inactivation.
Enzyme inactivation.
Water-activity reduction.
Recontamination prevention.

Finally separate:

- Physiological limit.
- Process parameter.
- Validation target.
- Legal maximum.
- Historical textbook value.

Master preservation matrix

Method	Principal target	Major surviving concern
Chilling	Growth rate	Psychrotrophs and prolonged shelf-life growth
Freezing	Growth and reaction rate	Survival, freeze-thaw damage and thaw-stage growth
Blanching	Enzymes and surface flora	Spores, residual enzymes and recontamination
Pasteurisation	Vegetative pathogens	Spores, heat-stable toxins and post-process contamination
UHT	Vegetative cells and many spores	Aseptic-package failure and post-opening growth
Retorting	Spores and vegetative cells	Underprocessing, cold spots and seam failure
Drying	aw-dependent growth	Survival and toxin persistence
Hurdle technology	Multiple homeostatic systems	Adaptation, local high-aw niches and hurdle interaction failure
Organic acids	Membrane and intracellular pH	Acid-tolerant organisms and pH-dependent loss of activity
Nisin	Susceptible Gram-positive cells	Gram-negative organisms, resistance and matrix inactivation
Natamycin	Fungal membranes	Inadequate surface coverage and penetration limits

Method	Principal target	Major surviving concern
Irradiation	DNA and cellular structures	Dose non-uniformity, resistant spores and recontamination

Final statutory rule

Never present an approximate physiological value, historical course value or international benchmark as a current Indian statutory requirement without checking the applicable FSSAI regulation, product category, amendment and notification.

End of Unit 2 Modules 3–8.